

SURV 626: Sampling

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Unit 1: Preliminaries

Course Description

Content

- Design vs. Analysis
 - Techniques (SRS, stratification, clustering, etc.)
 - Sampling variance estimation
- Finite population
- Design-based vs. Model-based inference
- Descriptive vs. Analytic
- Statistics
 - Means, proportions, totals
 - Variances
 - Correlations, regression coefficients

Unit 1: Preliminaries

Course Perspectives

- A. Design
- B. Typology & vocabulary
- C. Notation
- D. Sampling distribution

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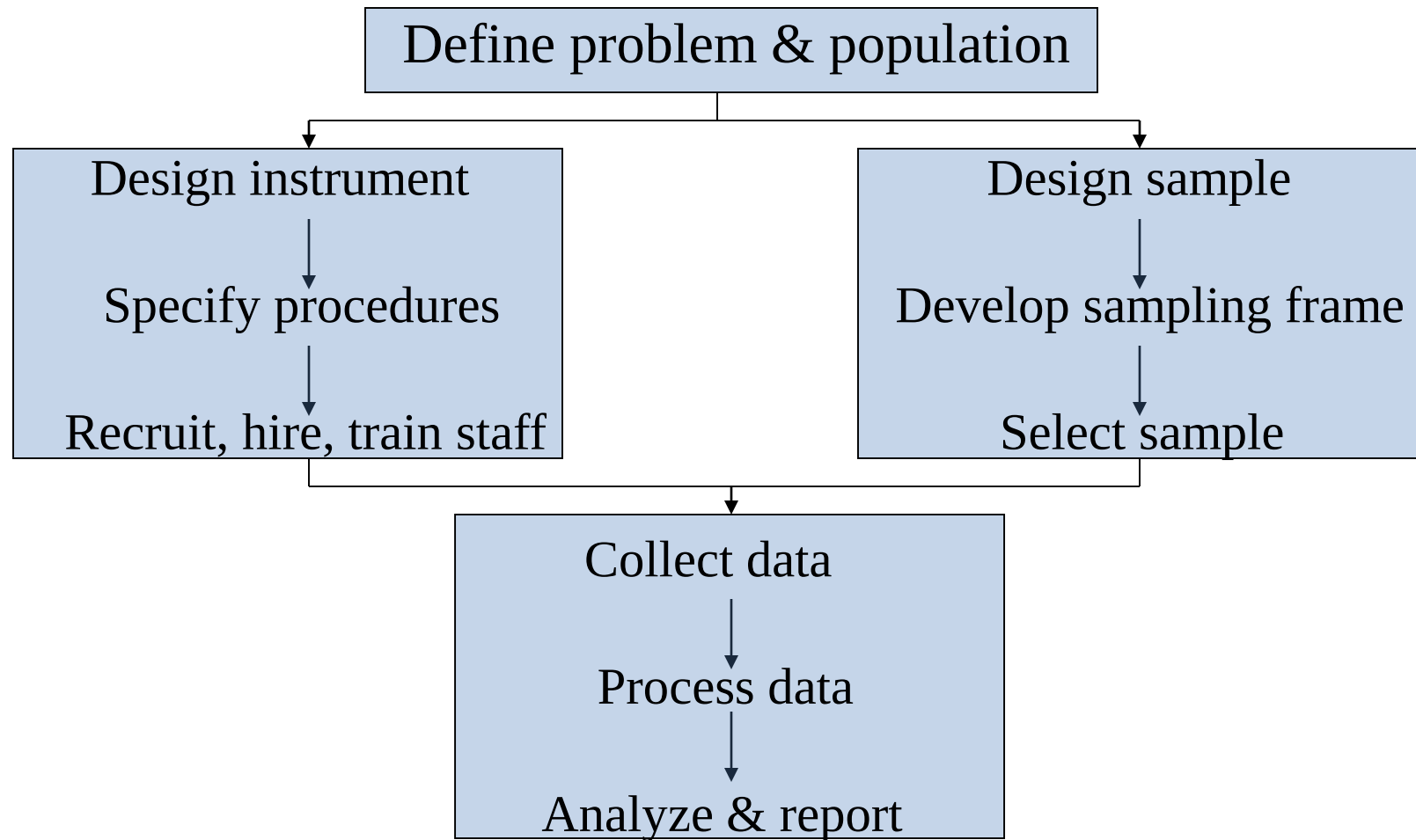
A. Design

- Research design
 - Experiments
 - Dependent variables
 - Factors
 - Control or randomization of disturbing variables
 - Quasi-experimental: observational
 - Survey samples: observational
- Design characteristics
 - Realism
 - Randomization
 - “Representation”

Sampling designs

- Probability sampling designs
 - Simple random sampling
 - Stratified sampling
 - Cluster sampling
 - Stratified cluster sampling, stratified multistage sampling
- Non-probability sampling designs
 - Convenience, haphazard
 - Snowball, Respondent Driven Sampling (RDS)
 - Volunteer
 - Quota sampling

Sample Surveys



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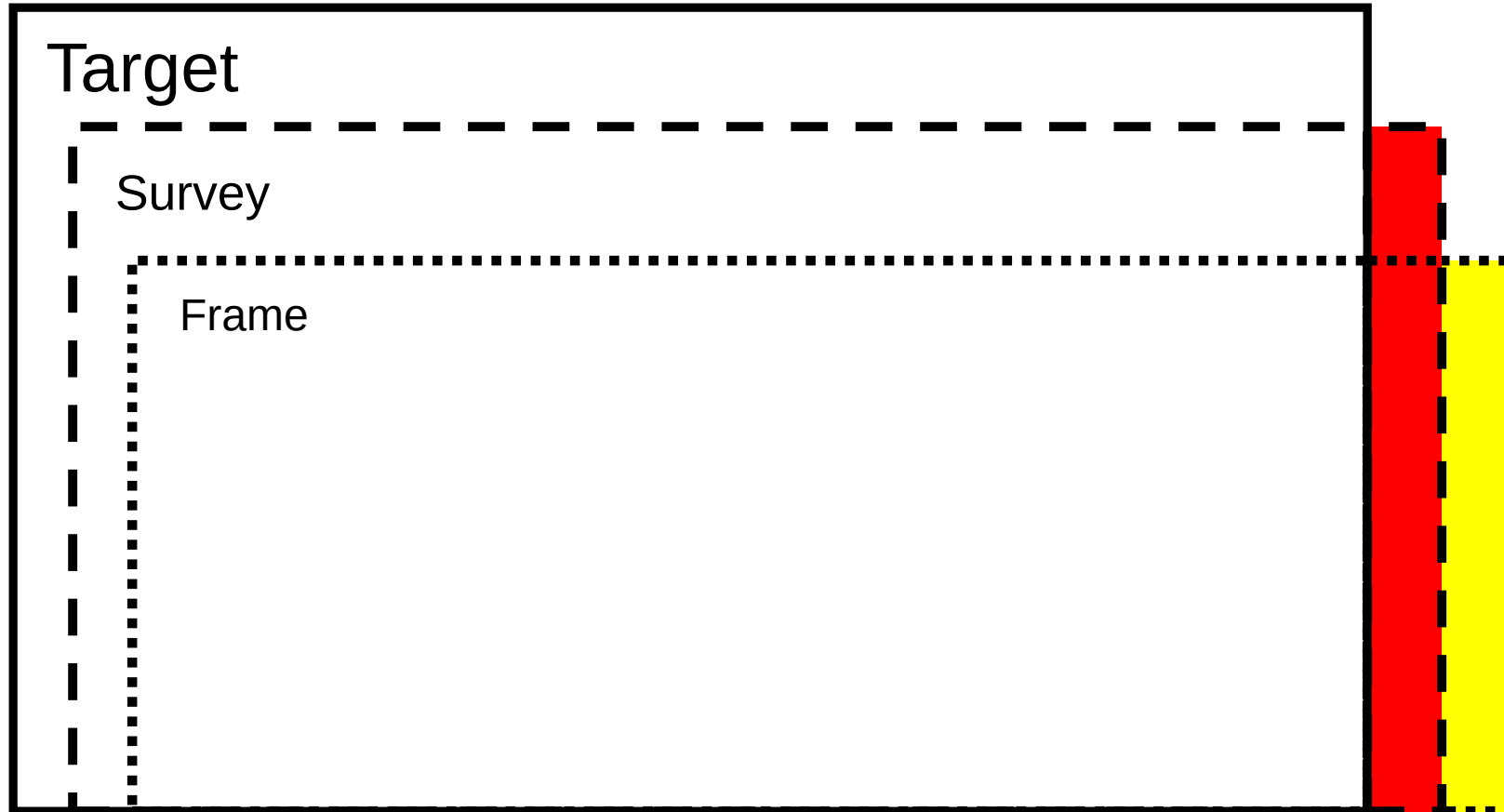
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B. Typology & vocabulary

- Finite population
- Large population & sample
- Randomization
- “Representative”
- Efficient (precision per unit cost)

Population



Survey Sampling Design Typology

Control	Unit	Chance	
		<i>Equal</i>	<i>Unequal</i>
<i>None</i>	<i>Element</i>	SRS or systematic	Non-epsem element
	<i>Cluster</i>	Cluster	Non-epsem cluster
<i>Grouped</i>	<i>Element</i>	Proportionate stratified random	Disproportionate stratified random
	<i>Cluster</i>	Proportionate stratified cluster	Disproportionate stratified cluster

Additional vocabulary

- Observation
 - Coverage
 - Response
- Quality
 - Reliability, variance
 - Validity, bias
- Properties
 - Unbiasedness
 - Precision
 - Accuracy (mean square error)
- Sampling distribution
 - Expectation
 - Standard error
- Inference
 - Central limit theorem
 - Confidence interval
 - Normal & t distributions
- Non-probability sampling
 - Volunteer
 - Purposive, judgment
 - Quota
- Measurability
- Sample design
 - Plan
 - Estimation

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C. Notation: Population

- N elements labels $i = 1, 2, \dots, N$
- Characteristics denoted $Y_1, Y_2, \dots, Y_N$

$$Y = Y_1 + Y_2 + \dots + Y_N$$

$$\bar{Y} = Y/N$$

$$P = Y/N$$

$$S^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i - \bar{Y})^2$$

$$S^2 = \left(\frac{1}{N-1} \right) \left(\sum_{i=1}^N Y_i^2 - Y^2/N \right)$$

$$S^2 = \left(\frac{N}{N-1} \right) P(1-P)$$

Sample

- Sample n elements from N , $i = 1, 2, \dots, n$
- Values denoted as $y_1, y_2, \dots, y_n$

$$y = y_1 + y_2 + \dots + y_n$$

$$\bar{y} = y/n$$

$$p = y/n$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$$

$$s^2 = \left(\frac{1}{n-1} \right) \left(\sum_{i=1}^n y_i^2 - y^2/n \right)$$

$$s^2 = \left(\frac{n}{n-1} \right) p(1-p) \approx p(1-p)$$

$$f = n/N \text{ and } F = f^{-1} = N/n$$

Unit 1: Preliminaries

Course Perspectives

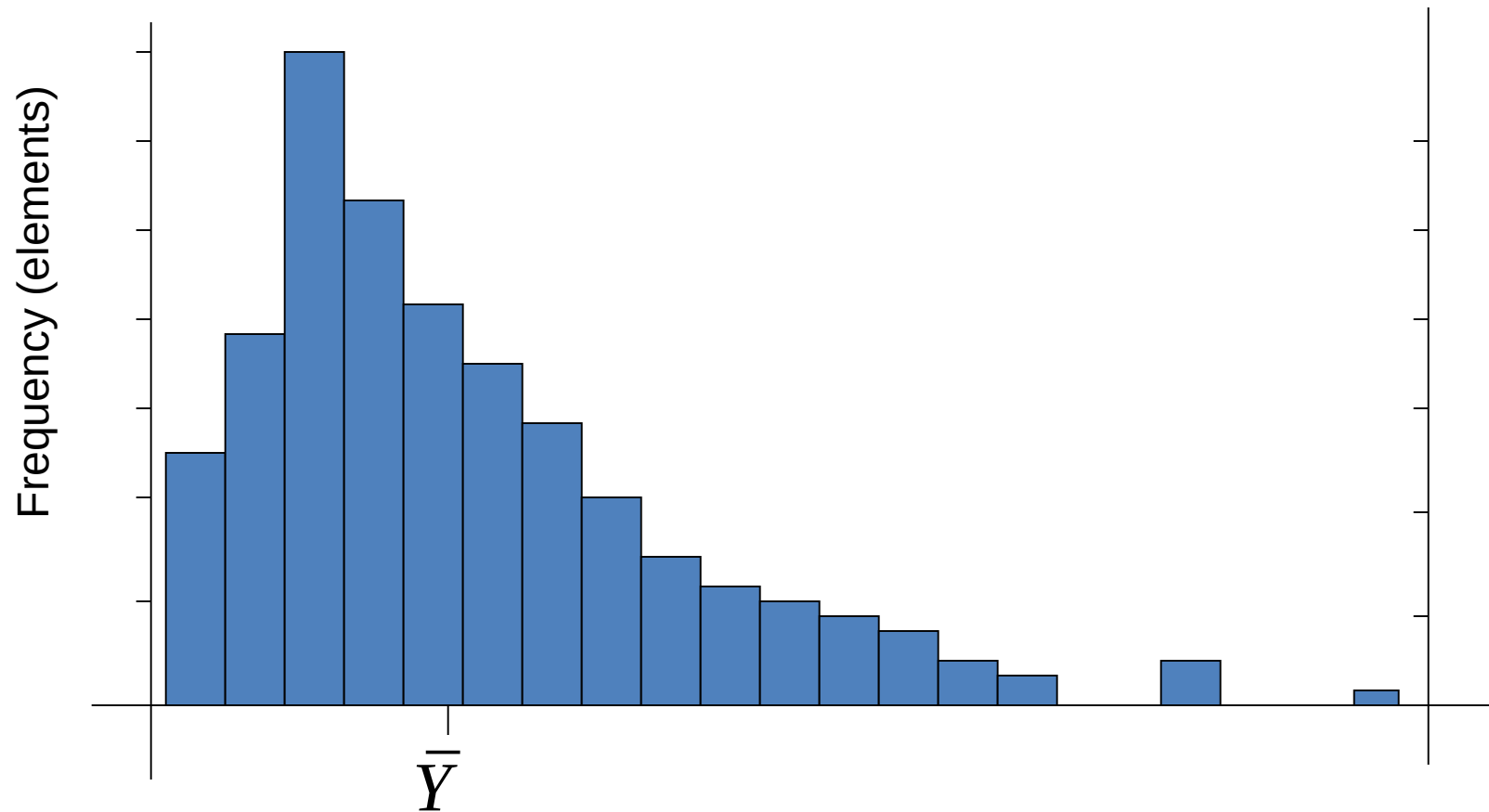
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D. Population distribution



Sampling distribution

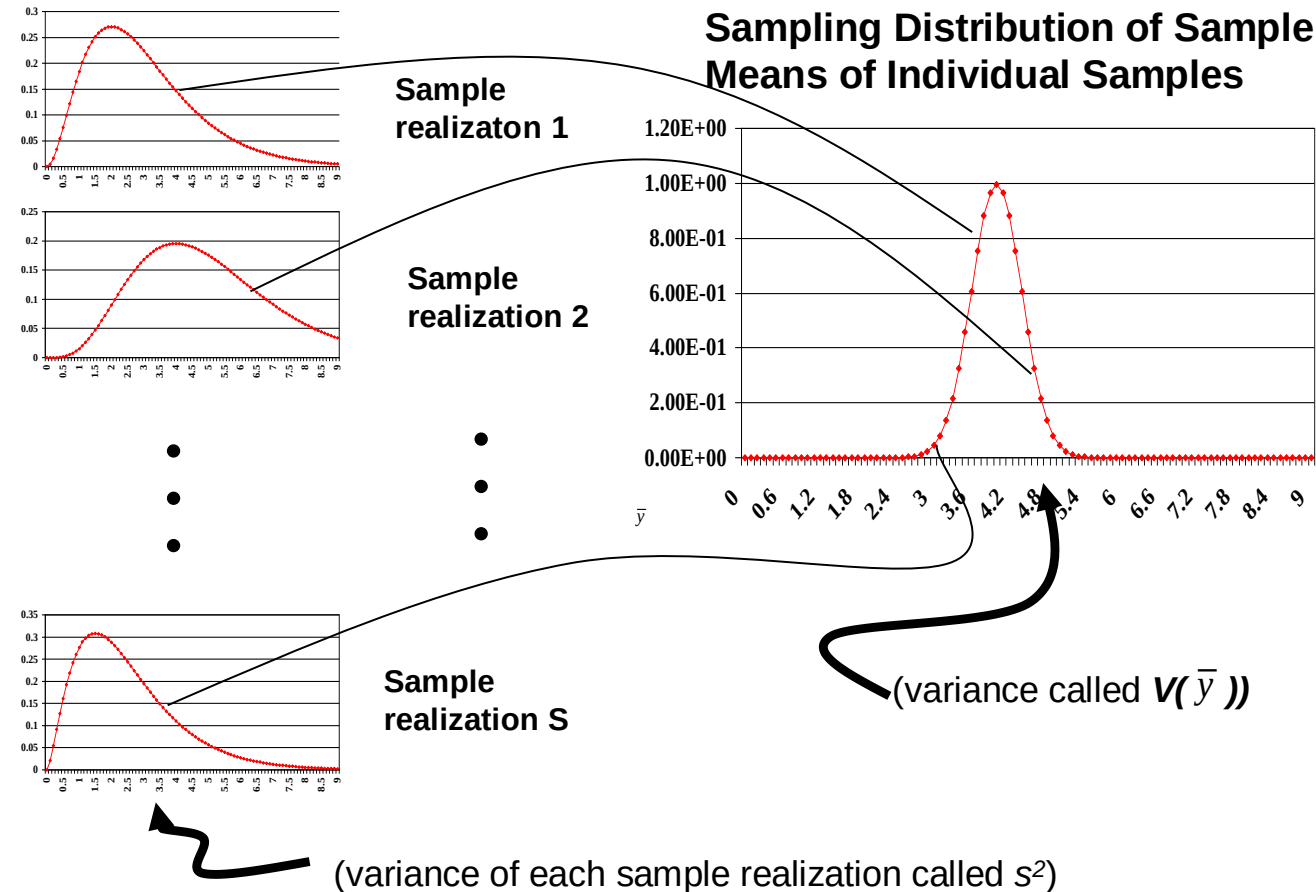


Figure 4.2 Distributions of y Variable from Sample Realizations Samples and the Sampling Distribution of the Mean

Unit 2: Elements Simple Random Sampling

- A. Implementation
- B. Precision & inference
- C. Sample size

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- A. Implementation
- B. Precision & inference
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A. Implementation

- Population of N , sample of n : $\binom{N}{n} = \frac{N!}{n!(N-n)!}$
- Create all samples of size n from N , select one at random
- Alternatively, random selection using mechanical device
 - Table of random numbers, using labels i to identify selected units

Seq. No.	Division	Sex	Ran	Salary
1	Eng&Prof	m	3	\$88
2	Medicine	f	3	\$45
3	Medicine	m	3	\$57
4	Medicine	m	1	\$133
5	Eng&Prof	f	2	\$71
6	Lit&SocSci	m	1	\$113
7	Medicine	f	3	\$65
8	Biol&Sci	m	3	\$47
9	Medicine	f	3	\$39
10	Lit&SocSci	m	1	\$74
11	Medicine	m	1	\$88
12	Lit&SocSci	m	1	\$62
13	Lit&SocSci	m	1	\$49
14	Medicine	m	3	\$88
15	Medicine	m	1	\$181
16	Eng&Prof	m	3	\$63
17	Medicine	m	2	\$94
18	Eng&Prof	m	1	\$91
19	Medicine	m	1	\$60
20	Eng&Prof	m	3	\$55
21	Biol&Sci	m	2	\$55
22	Medicine	f	1	\$106
23	Medicine	m	1	\$116
24	Medicine	m	3	\$79
25	Lit&SocSci	m	1	\$61
26	Lit&SocSci	f	3	\$37
27	Medicine	m	2	\$72
28	Eng&Prof	m	1	\$105
29	Medicine	m	2	\$79
30	Medicine	m	1	\$61
31	Medicine	m	1	\$86
32	Biol&Sci	m	1	\$103
33	Lit&SocSci	m	1	\$48
34	Eng&Prof	m	2	\$64
35	Eng&Prof	m	1	\$78
36	Medicine	f	2	\$53
37	Biol&Sci	m	1	\$85
38	Eng&Prof	m	1	\$61
39	Medicine	m	1	\$106
40	Lit&SocSci	m	2	\$60
41	Biol&Sci	f	1	\$73
42	Medicine	m	1	\$70
43	Medicine	f	3	\$32
44	Lit&SocSci	m	2	\$49
45	Eng&Prof	m	3	\$43
46	Medicine	m	1	\$75
47	Lit&SocSci	m	1	\$92
48	Medicine	m	2	\$107
49	Biol&Sci	m	2	\$57
50	Medicine	m	2	\$114
51	Eng&Prof	m	3	\$55
52	Biol&Sci	m	1	\$49

53	Eng&Prof	m	3	\$57
54	Medicine	m	1	\$118
55	Medicine	m	3	\$84
56	Eng&Prof	m	3	\$52
57	Medicine	m	3	\$64
58	Eng&Prof	m	1	\$75
59	Medicine	f	1	\$87
60	Eng&Prof	m	3	\$58
61	Medicine	f	3	\$39
62	Medicine	m	3	\$69
63	Medicine	f	2	\$46
64	Eng&Prof	f	1	\$86
65	Medicine	m	3	\$87
66	Medicine	m	3	\$59
67	Eng&Prof	f	3	\$44
68	Medicine	m	2	\$123
69	Lit&SocSci	f	3	\$37
70	Lit&SocSci	m	1	\$106
71	Lit&SocSci	m	1	\$91
72	Lit&SocSci	m	1	\$78
73	Biol&Sci	m	1	\$77
74	Medicine	m	1	\$90
75	Eng&Prof	m	2	\$71
76	Medicine	f	3	\$42
77	Medicine	f	2	\$59
78	Eng&Prof	m	2	\$49
79	Biol&Sci	m	1	\$83
80	Lit&SocSci	m	1	\$34
81	Medicine	f	3	\$42
82	Medicine	m	2	\$97
83	Medicine	m	1	\$109
84	Lit&SocSci	f	2	\$48
85	Medicine	m	1	\$47
86	Eng&Prof	m	2	\$45
87	Medicine	m	3	\$83
88	Medicine	m	2	\$51
89	Biol&Sci	m	1	\$78
90	Lit&SocSci	m	1	\$70
91	Eng&Prof	f	2	\$46
92	Eng&Prof	m	1	\$85
93	Lit&SocSci	m	1	\$53
94	Medicine	f	3	\$40
95	Eng&Prof	m	1	\$87
96	Lit&SocSci	m	1	\$71
97	Medicine	m	1	\$75
98	Biol&Sci	m	1	\$85
99	Lit&SocSci	m	2	\$50
100	Medicine	m	3	\$118
101	Lit&SocSci	m	2	\$55
102	Medicine	m	3	\$80
103	Eng&Prof	m	1	\$114
104	Lit&SocSci	m	1	\$63
105	Medicine	m	1	\$112

Table of Random Numbers

	1-5	6-10	11-15	16-20	21-25	26-30	31-35	36-40	41-45	46-50
1	57982	03250	09488	12085	38325	01091	03826	77648	57881	04677
2	34104	88576	32549	09603	73860	43758	05470	38344	04492	10730
3	01943	29119	72315	70334	98029	80936	37792	45017	18819	84787
4	25329	02138	52670	11692	03654	91109	17357	27013	32767	38642
5	23869	85541	29993	57693	84597	43545	23014	06089	52569	93160
6	69419	03843	33691	19541	80307	96350	62760	07467	12970	83870
7	14078	76539	08538	17888	65481	03995	63362	29308	23196	51186
8	14339	07381	13958	46562	75860	66994	97734	83135	77941	39411
9	97670	49015	97498	01260	63189	29044	11089	81803	87889	81923
10	15279	11006	82780	74184	34105	09246	31268	91081	99324	68381
11	97947	01993	12603	63485	16937	06190	77536	18598	92322	21257
12	00631	55073	73183	05961	16402	39801	96560	88053	70254	73836
13	30246	12514	24792	46736	61024	87174	63045	35815	57531	41129
14	74016	75832	78113	18374	78851	46324	78629	54165	73712	08415
15	53669	78559	66231	29085	78506	80934	88520	86912	17774	29711
16	98280	13316	23734	63720	50355	74201	17325	69300	28342	91793
17	32396	30527	10030	69065	87877	81658	23343	87272	74683	86065
18	05708	99540	11289	09030	94001	05587	41643	40782	25630	48366
19	04004	49911	47941	81165	14505	07432	79939	01309	79165	57871
20	84424	61863	81655	87420	76149	74712	81555	82645	05850	54779
21	84546	16063	09051	31787	57235	27302	83322	48361	11630	80174
22	76882	00300	45868	17098	53385	83121	70986	70598	91543	44137
23	98169	25789	99043	73890	35327	91723	77932	77971	41355	19562
24	33621	78638	27009	29610	20171	74951	34482	89391	81210	86209
25	61507	79038	38331	70320	73347	57565	19920	41574	68935	79233
26	20920	81284	01289	66936	60782	71442	63257	45294	11141	15333
27	78861	84019	31418	64943	42169	88995	48423	95960	27581	11679
28	08307	99947	51173	87234	60693	06143	13459	59514	79507	38474
29	99441	02441	66284	05523	19477	51695	58978	92845	69511	81076
30	79407	56191	54855	83823	33322	96478	39901	82641	38556	56564
31	34643	05408	67404	45254	37951	12189	76651	51762	59986	83559
32	79071	00901	91791	64053	47239	64456	10573	91720	32152	70961
33	39776	75808	19351	39680	75676	29758	11181	48497	06845	08877
34	39404	45516	97619	49788	96815	22561	01201	50766	19787	53701
35	48094	09146	14671	56419	47127	53938	80707	35195	31953	27177
36	18410	40494	13931	61713	27618	34620	72264	38473	21435	08084
37	67160	05126	99743	18590	94419	32835	22290	76000	81239	43728
38	08555	37978	24194	23612	83749	24952	05356	66661	57819	99209
39	68035	97917	09711	82240	05453	43354	44223	19470	20455	93951
40	12989	19518	39150	20121	60930	48430	29572	66745	10380	89898
41	54634	90880	59787	00767	25687	88036	32752	27228	52429	14533
42	22603	31768	32309	68558	90881	87299	30044	06782	43324	65189
43	59743	43696	26720	27716	39721	74953	90138	90812	86546	07429
44	97687	86485	58473	00533	12837	74736	31091	42786	44844	10143
45	45106	28013	06476	46786	61360	15114	00315	45583	85248	59929
46	05655	11433	54983	86786	26287	75827	57443	70597	38351	91876
47	51610	48745	02202	10037	94816	87585	15188	64636	70648	50783
48	86277	69022	37506	91519	22285	04022	49127	77337	93702	80773
49	51243	81450	22865	61165	40356	58407	82449	40460	28950	00041
50	72699	19280	12900	57824	58612	00859	35035	82783	15422	45453

A. Implementation

- Population of N , sample of n : $\binom{N}{n} = \frac{N!}{n!(N-n)!}$
- Create all samples of size n from N , select one at random
- Alternatively, random selection using mechanical device
 - Table of random numbers, using labels i to identify selected units
 - Software using pseudo-random numbers
 - Not haphazard
- Base procedure
 - SRS – without replacement
 - SRS seldom used in practice, even from a simple frame
 - All samples of n distinct elements equally likely

Probabilities of selection

- Draw by draw ...
- With replacement (Unrestricted Random, URS):
 - $p_{ij} = 1/N$, for draw j and element $i = 1, \dots, N$
- For SRS, across all draws, $\pi_i = f = n/N$:

$$p_1 = \frac{1}{N}$$

$$p_2 = (1 - p_1) \left[\frac{1}{N-1} \right] = \left[\frac{N-1}{N} \right] \left[\frac{1}{N-1} \right] = \frac{1}{N}$$

$$p_3 = (1 - p_1)(1 - p_2) \left[\frac{1}{N-2} \right] = \frac{1}{N}$$

Sample estimates

$$y_1, y_2, \dots, y_n \quad y_1 + y_2 + \dots + y_n = \sum_{i=1}^n y_i = y$$

$$\bar{y} = \sum_{i=1}^n y_i / n = y/n$$

$$s^2 = \left[(y_1 - \bar{y})^2 + \dots + (y_n - \bar{y})^2 \right] / (n-1)$$

$$= \sum_{i=1}^n (y_i - \bar{y})^2 / (n-1)$$

$$= \left(\sum_{i=1}^n y_i^2 - \frac{y^2}{n} \right) / (n-1)$$

Unit 2: Elements Simple Random Sampling

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B. Precision & inference

$$\bar{y} = y/n$$

$$Var(\bar{y}) = \frac{1}{\binom{N}{n}} \sum_{s=1}^{\binom{N}{n}} (\bar{y}_s - \bar{Y})^2 = \frac{(1-f)}{n} S^2$$

$$\begin{aligned} \text{var}(\bar{y}) &= \frac{(1-f)}{n} s^2 = \\ &= \frac{(1-f)}{n} \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{(n-1)} = \frac{\left(1 - \frac{n}{N}\right)}{n} \frac{\left(\sum_{i=1}^n y_i^2 - \frac{y^2}{n}\right)}{(n-1)} \end{aligned}$$

$$se(\bar{y}) = \sqrt{\text{var}(\bar{y})}$$

Other estimates: Population Total

$$\hat{Y} = Fy = \left(\frac{N}{n}\right)y = N\bar{y}$$

$$Var(\hat{Y}) = F^2 Var(y) = N^2 Var(\bar{y})$$

$$var(\hat{Y}) = N^2 var(\bar{y}) = N^2 \frac{(1-f)}{n} s^2$$

$$se(\hat{Y}) = N \sqrt{var(\bar{y})}$$

Other estimates: Proportions

- Values of Y are zero or 1:

$$p = \bar{y}$$

$$Var(p) = \frac{(1-f)}{n} S^2 = \frac{(1-f)}{n} \left(\frac{N}{N-1} \right) P(1-P)$$

$$var(p) = \frac{(1-f)}{n} s^2 = \frac{(1-f)}{n} \left(\frac{n}{n-1} \right) p(1-p) = \frac{(1-f)}{n-1} p(1-p)$$

Inference

- Standard deviation & element variance

$$s = \sqrt{s^2}$$

- Standard error & sampling variance

$$se(\bar{y}) = \sqrt{\text{var}(\bar{y})} = \sqrt{\frac{(1-f)}{n} s^2} = s \sqrt{\frac{(1-f)}{n}}$$

- Confidence interval

$$\bar{y} \pm t_{1-\alpha/2, n-1} \sqrt{\text{var}(\bar{y})} = \bar{y} \pm t_{1-\alpha/2, n-1} \sqrt{\frac{(1-f)}{n} s^2}$$

$$p \pm t_{1-\alpha/2, n-1} \sqrt{\text{var}(p)} = p \pm t_{1-\alpha/2, n-1} \sqrt{\frac{(1-f)}{n-1} p(1-p)}$$

Without Replacement Sampling

- Population of $N = 5$, **without** replacement selection of $n = 3$:

(1) Y_1, Y_2, Y_3 $\bar{y}_1$

(2) Y_1, Y_2, Y_4 $\bar{y}_2$

•

•

•

(10) Y_3, Y_4, Y_5 $\bar{y}_{10}$

$$Var(\bar{y}) = \frac{1}{10} \sum_{s=1}^{10} \left(\bar{y}_{(s)} - \bar{Y} \right)^2 = \frac{(1-f)}{n} S^2$$

With Replacement Sampling

- Population of $N = 5$, **with** replacement selection of $n = 3$:

(1) Y_1, Y_1, Y_1 $\bar{y}_1$

(2) Y_1, Y_1, Y_2 $\bar{y}_2$

•

•

•

(125) Y_5, Y_5, Y_5 $\bar{y}_{125}$

$$Var(\bar{y}) = \frac{1}{125} \sum_{s=1}^{125} \left(\bar{y}_{(s)} - \bar{Y} \right)^2 = \frac{1}{n} S^2$$

Without or With?

- Population of $N = 5$, **without** and **with** replacement selection of $n = 3$:

(1) Y_1, Y_2, Y_3

(2) Y_1, Y_2, Y_4

•

•

•

(10) Y_3, Y_4, Y_5

$$Var(\bar{y}) = \frac{(1-f)}{n} S^2$$

(1) Y_1, Y_1, Y_1

(2) Y_1, Y_1, Y_2

•

•

•

(125) Y_5, Y_5, Y_5

$$Var_{URS}(\bar{y}) = \frac{1}{n} S^2$$

Proportions: useful properties

$$s^2 = \left(\frac{n}{n-1} \right) p(1-p) \approx p(1-p)$$

$$\text{var}(p) = \frac{(1-f)}{(n-1)} p(1-p)$$

P	$S^2 \approx P \times Q$	$S \approx \sqrt{P \times Q}$
0.50	0.250	0.500
0.60	0.240	0.490
0.70	0.210	0.458
0.95	0.048	0.218

Proportions: useful properties

- What does this mean for us?
- Projection: estimating standard errors before we draw a sample
 - For proportions, p can project s without large error

p	$s^2 \approx p \times q$	$s \approx \sqrt{p \times q}$
0.50	0.250	0.500
0.60	0.240	0.490
0.70	0.210	0.458
0.95	0.048	0.218

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C. Sample size

- Sampling error depends very little on N
 - Unless f greater than 0.05
 - (Approximately) Same precision for $n = 1,000$ in Ann Arbor or the People's Republic of China
- Sample size determination in SRS
 - Specify desired level of precision
 - Obtain value for element variance
 - Solve for n

Computation

- Desired precision: $\hat{V}^2 = \left(1 - \frac{n}{N}\right) \frac{S^2}{n}$
- Solve algebraically for n : $n = \frac{\hat{S}^2}{\hat{V}^2 + \frac{\hat{S}^2}{N}}$
- Ignoring the *fpc*: $n = \frac{\hat{S}^2}{\hat{V}^2}$

Desired precision: confidence interval or margin of error (e)

- The desired level of precision can also be specified by the width of a confidence interval:

$$(L, U) = \left(\bar{y} - t_{1-\alpha/2, n-1} \sqrt{\text{var}(\bar{y})}, \bar{y} + t_{1-\alpha/2, n-1} \sqrt{\text{var}(\bar{y})} \right)$$

- Then

$$t_{1-\alpha/2, n-1} \sqrt{\text{var}(\bar{y})} = \left(\frac{U - L}{2} \right) = e$$

$$\hat{V} = \sqrt{\text{var}(\bar{y})} = \frac{U - L}{2t_{1-\alpha/2, n-1}} = \frac{e}{z_{1-\alpha/2}}$$

Alternatively...

- Desired precision: $\hat{V}^2 = \left(\frac{e}{z_{1-\alpha/2}} \right)^2 = \left(1 - \frac{n}{N} \right) \frac{S^2}{n}$
- Solve algebraically for n : $n = \frac{\hat{S}^2}{\left(\frac{e}{z_{1-\alpha/2}} \right)^2 + \frac{\hat{S}^2}{N}}$
- Ignoring the *fpc*: $n = \frac{s^2}{\left(\frac{e}{z_{1-\alpha/2}} \right)^2}$

Relative Precision Alternative

- $cv(\bar{y}) = \frac{se(\bar{y})}{\bar{y}} = \frac{\hat{V}}{\bar{y}}$
- For example, supposed that we want $cv(\bar{y}) = 0.05$ for $\bar{y} = p = 0.1$

$$\hat{V} = se(\bar{y}) = \bar{y} \times cv(\bar{y}) = 0.1 \times 0.5 = 0.005$$

- Then, calculate sample size as on previous slide:

$$n = \frac{\hat{S}^2}{\hat{V}^2 + \frac{\hat{S}^2}{N}}$$

Example: adjusting a preliminary n

$$n = \frac{\hat{S}^2}{\hat{V}^2 + \frac{\hat{S}^2}{N}}$$

$$\hat{S}^2 = 6,250, \hat{V} = 2.5$$

$$N = 100,000$$

$$n = \frac{6,250}{2.5^2 + \frac{6,250}{100,000}} = 991$$

$$n' = \frac{\hat{S}^2}{\hat{V}^2}, \quad n = \frac{n'}{1 + \frac{n'}{N}}$$

$$\hat{S}^2 = 6,250, \hat{V} = 2.5$$

$$n' = \frac{\hat{S}^2}{\hat{V}^2} = \frac{6,250}{2.5^2} = 1,000$$

$$N = 100,000$$

$$n = \frac{n'}{1 + \frac{n'}{N}} = \frac{1,000}{1 + 0.1} = 991$$

Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
- C. Blanks
- D. Duplicates
- E. Clustering
- F. Many-to-many matching

Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
- C. Blanks
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- E. Clustering
- F. Many-to-many matching

A. Frames

- Frame: **set of materials and/or procedures** used to designate a sample of units
- Rule links frame elements to population elements
- Accurate and up-to-date frames located in one location preferred
- Numbered, computerized lists are best.

Unit 2: Elements

Frame Problems

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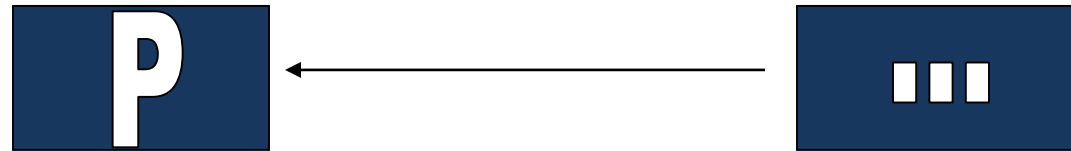
Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
- C. Blanks
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- E. Clustering
- F. Many-to-many matching

B. Non-coverage

- Some population elements are not on the frame
 - Have zero chance of selection.



- Solution: use supplemental frames that cover non-covered elements
- Solution: use non-coverage weighting adjustments
 - Considered under post-stratification

B. Non-coverage

Target Population
Elements

Frame Population
Elements

425 Kingsley St.	←	314-555-9124
3737 Geddes Rd.		
45 S. Main, Apt. 2	←	314-555-9156
551 Seventh St.	←	314-555-8847
1845 W. Liberty	←	314-555-9173
968 Hill St.	←	314-555-9222
399 N. Fourth Ave.	←	314-555-9274
.		.
.		.
.		.

**Target Population Element Without
a Sampling Frame Element**

Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
- C. Blanks
- D. Duplicates
- E. Clustering
- F. Many-to-many matching

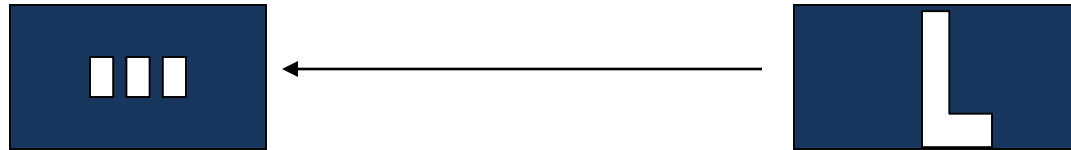
Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
- C. **Blanks**
- D. Duplicates
- E. Clustering
- F. Many-to-many matching

C. Blanks

- Frame elements do not have corresponding population elements:



- Know frame element is blank after selection
- Screening to find eligible list elements
- One solution to **avoid**: substitute the next listing

C. Blanks

Target Population
Elements

Frame Population
Elements

425 Kingsley St.	←—————	314-555-9124
(Not in service)	←-----	314-555-1433
45 S. Main, Apt. 2	←—————	314-555-9156
551 Seventh St.	←—————	314-555-8847
1845 W. Liberty	←—————	314-555-9173
968 Hill St.	←—————	314-555-9222
399 N. Fourth Ave.	←—————	314-555-9274
.		.
.		.
.		.

**No Target Population Element for
a Sampling Frame Element**

Rejecting Foreign Elements

- Solution: reject blank
 - Leads to smaller sample size than desired
 - Adjust the sampling rate (increase number of listings):
 - Let p_b denoted estimated proportion blank elements
 - n is the desired final sample size of elements (and not list elements)
 - m is the number of frame elements to select.

$$m = \frac{n}{(1 - p_b)}$$

Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
- C. Blanks
- D. Duplicates
- E. Clustering
- F. Many-to-many matching

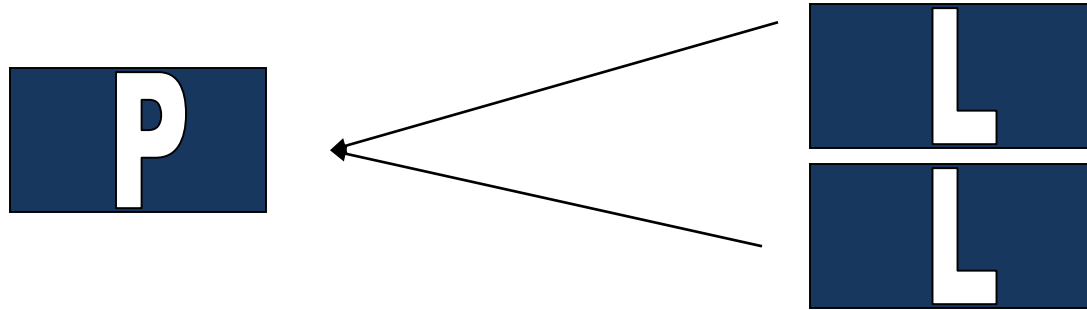
Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
- C. Blanks
- D. Duplicates**
- E. Clustering
- F. Many-to-many matching

D. Duplicates

- Occur when a single population element is linked to two or more frame elements

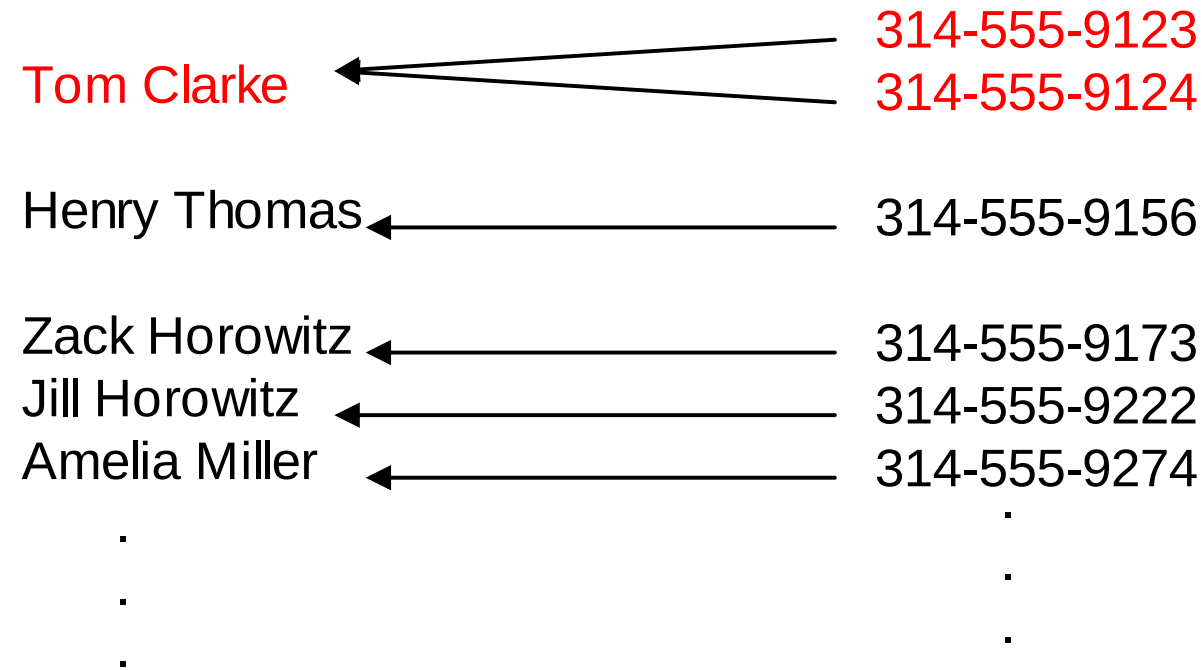


- If only a few readily identified, remove from the list before selection
- Eliminating duplicates from the sample still leaves unequal probabilities of selection

D. Duplicates

Target Population
Elements

Frame Population
Elements



**Duplication of Target Population Elements
By More Than One Sampling Frame Element**

Other solutions

- Choose unique listing
 - First, last, largest, or randomly chosen frame listing
- Determine how many duplicates for a given selected element
 - Say d_i .
- Weight observation for that selection with probability $1/d_i$.

Duplicate Weighting Illustration

- *Epsem*: $f = n/N$ for $i = 1, 2, \dots, N$
- The unweighted mean is appropriate:

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} = \frac{y_1 + y_2 + \dots + y_n}{1+1+\dots+1}$$

- For example, from $N = 2,000$ families select $n = 25$ with *epsem*

$$f = \frac{25}{2000} = \frac{1}{80} \longrightarrow w_i = f^{-1} = 80$$

Weighted Means

- *Epsem*: $f = n/N = 1/80$ for $i = 1, 2, \dots, N$
 - The weighted mean is

$$\bar{y} = \bar{y}_w = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i} = \frac{80y_1 + 80y_2 + \dots + 80y_n}{80 + 80 + \dots + 80} = \frac{\sum_{i=1}^n y_i}{n}$$

Duplicates: Non-*epsem* Illustration

- Non-*epsem* sampling has different selection probability for elements
- For example, suppose that $n = 25$ children are selected from a list of $N = 4,000$ children enrolled in school
 - Estimating family characteristics
- For families with one child enrolled $d_i=1$

$$\pi_i = \frac{25}{4000} = \frac{1}{160} \longrightarrow w_i = \pi_i^{-1} = 160$$

Unequal Chances: Over-representation

- For families with two children enrolled $d_i=2$ and

$$\pi_i = 2 \times \frac{25}{4000} = \frac{1}{80} \longrightarrow w_i = \pi_i^{-1} = 80$$

- For families with three children enrolled $d_i=3$ and

$$\pi_i = 3 \times \frac{25}{4000} = \frac{1}{53.33} \longrightarrow w_i = \pi_i^{-1} = 53.33$$

Weighted Mean, Revisited

- Compensate for varying probabilities using the weighted estimator

$$\bar{y} = \bar{y}_w = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i} = \frac{(160) y_1 + (53.33) y_2 + \dots + (80) y_n}{160 + 53.33 + \dots + 80} \neq \frac{\sum_{i=1}^n y_i}{n}$$

Unit 2: Elements

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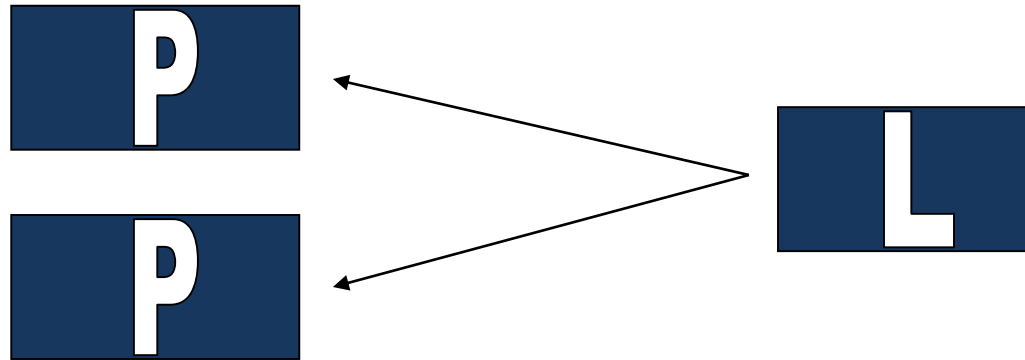
Unit 2: Elements

Frame Problems

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E. Clustering

- Occurs when more than one population element can be selected by a single frame element

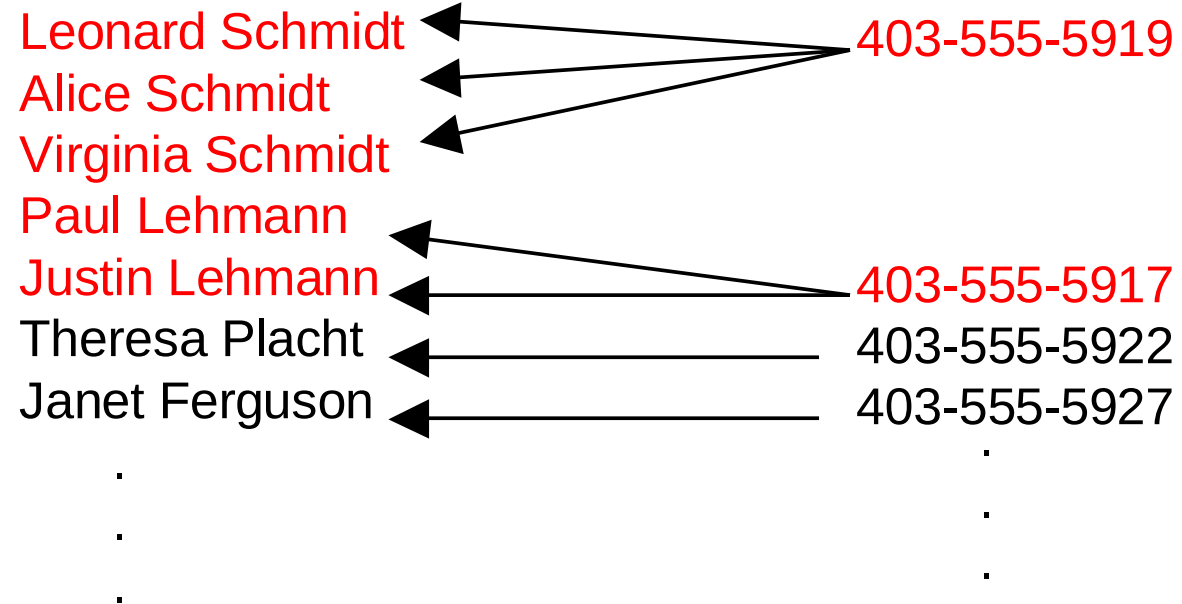


- Take all elements within selected clusters
 - Clusters unequal size: sample size varies
 - Probabilities of selection remain equal, and unweighted estimates remain unbiased

E. Clustering

Target Population
Elements

Frame Population
Elements



**Clustering Target Population
Elements Relative to Sampling Frame Elements**

Other solutions

- Take all elements in selected clusters, adjust the sample size in advance
- $m = n \times \bar{b}$
- If clustering is frequent, use cluster sampling techniques with subsampling
 - If the clusters selected with equal chances (i.e., *epsem*), subsampled elements are not
 - Weight to compensate for unequal probabilities

Objective Respondent Selection

- In the social sciences, sampling a single element from small clusters of unequal size occurs often
 - Sampling individuals from households
 - Often randomized selections are carried out by computer
- Techniques for households:
 - Objective Respondent Selection proposed by Kish
 - Rizzo method
 - Trol Dahl-Carter method
 - *(1) How many people (18 or older) live in your household... counting yourself?*
 - *(2) How many of them are men?*
 - Nearest birthday method

Unit 2: Elements

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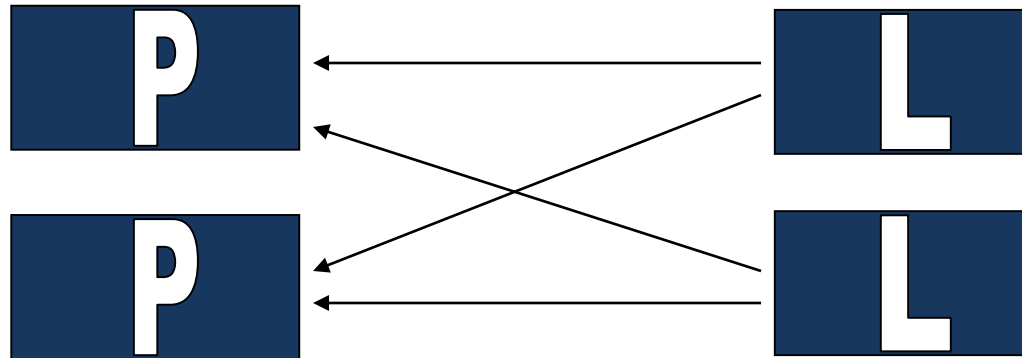
Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
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F. Many-to-Many Matching

- Occurs when more than one population element can be selected by more than one frame element

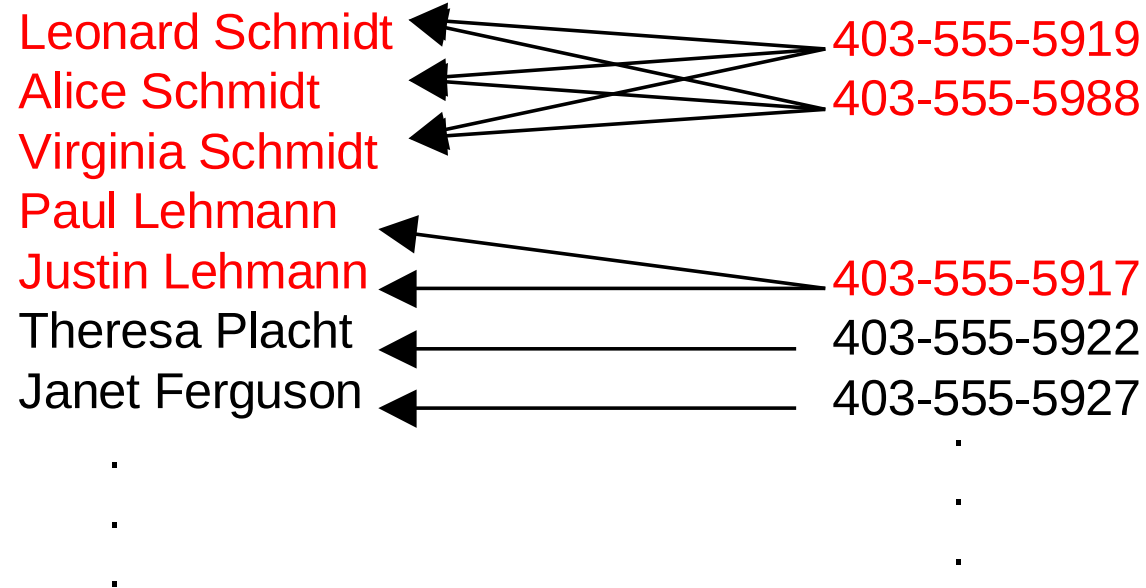


- Solution: combinations of weighting and subsampling often used

F. Many-to-Many Matching

Target Population
Elements

Frame Population
Elements



Clustering & Duplicates

Unit 3: Stratification

Stratified Random Sampling

- A. Purpose of stratification
- B. Notation
- C. Inference
- D. Precision & design effect

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A. Purpose of stratification

- Example: Estimate the mean energy consumption of commercial buildings in a state (no. of kW-h)
- Frame with $N = 5,318$ commercial buildings

Building ID	Location	Size	...
1	Area 1	580,200	
2	Area 1	45,630	
⋮	⋮	⋮	
5,318	Area 23	360,380	

- Objectives:
 - Control the distribution of sample buildings by size (square footage)
 - Improve the precision over SRS

SRS review

- Objective, randomized selection
- Every element has an equal chance of selection
- Every possible sample has the same probability of being drawn
- Even "bad" samples have an equal chance of being selected
 - “Bad samples”: a break down of the randomization principle
- Further, other sample designs can achieve smaller sampling error for the same cost
- Or same sampling error for lower cost

Stratified sampling

- Objective:
 - Can we sample to avoid, or even eliminate, "bad" samples?
 - Can we achieve smaller sampling error for the same cost or same sampling error for lower cost?
- Stratification uses information other than labels (i.e., $i = 1, 2, \dots, N$)
 - "Auxiliary information" used to create strata
 - Group elements into H mutually exclusive and exhaustive groups (strata)
 - Select samples from each stratum
 - Compute estimates for each stratum, and combine

Stratified sampling: Example

- Control distribution of the sample buildings by size (square footage):

Stratum	Group	Count
1	< 50K sq. ft.	1,182
2	50 - 250K	1,846
3	250 - 4,000K	1,395
4	>= 4,000K	895
Total		5,318

Unit 3: Stratification

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B. Notation

- Notation for stratified sampling is same as that for SRS, with a subscript to denote strata
- For the population N elements in H strata with N_h elements in stratum h :

- $$\sum_{h=1}^H N_h = N_1 + N_2 + \dots + N_H = N$$

- $$Y_{h1}, Y_{h2}, \dots, Y_{hN_h}, h = 1, \dots, H$$

Population level

- For each stratum h :

$$Y_h = \sum_{i=1}^{N_h} Y_{hi}$$

$$\bar{Y}_h = \frac{Y_h}{N_h} = \frac{1}{N_h} \sum_{i=1}^{N_h} Y_{hi}$$

$$S_h^2 = \frac{1}{N_h - 1} \sum_{i=1}^{N_h} (Y_{hi} - \bar{Y}_h)^2$$

- For dichotomous variables:

$$\bar{Y}_h = P_h \quad S_h^2 = \frac{N_h}{N_h - 1} P_h (1 - P_h)$$

Sample level

- Sample sizes n_h and sample elements $y_{h1}, y_{h2}, \dots, y_{hn_h}$:

$$y_h = \sum_{i=1}^{n_h} y_{hi}$$

$$\bar{y}_h = \frac{y_h}{n_h} = \frac{1}{n_h} \sum_{i=1}^{n_h} y_{hi}$$

$$s_h^2 = \frac{1}{n_h - 1} \sum_{i=1}^{n_h} (y_{hi} - \bar{y}_h)^2$$

$$\sum_{h=1}^H n_h = n_1 + n_2 + \dots + n_H = n$$

- For dichotomous variables:

$$\bar{y}_h = p_h \quad s_h^2 = \frac{n_h}{n_h - 1} p_h (1 - p_h)$$

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	n_h	$\bar{y}_h$	s_h^2
1	< 50K sq. ft.	1,182	100	72.7	11,783.6
2	50 - 250K	1,846	200	336.1	205,657.8
3	250 - 4,000K	1,395	120	1,690.2	3,568,970.7
4	$\geq 4,000K$	895	80	11,659.0	183,039,437.2
Total		5,318	500		

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Unit 3: Stratification

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C. Inference

- At population level,

$$\bar{Y} = \frac{\sum_{h=1}^H \sum_{i=1}^{N_h} Y_{hi}}{N} = \frac{\sum_{h=1}^H Y_h}{N} = \frac{\sum_{h=1}^H \left(\frac{N_h}{N} \right) Y_h}{N} = \frac{\sum_{h=1}^H N_h \left(\frac{Y_h}{N_h} \right)}{N} =$$

$$\sum_{h=1}^H \frac{N_h}{N} \bar{Y}_h = \sum_{h=1}^H W_h \bar{Y}_h$$

- At sample level,

$$\bar{y}_w = \sum_{h=1}^H W_h \bar{y}_h$$

Element level weighting

- The stratum level weighted mean can be expressed with element level weights:

$$\begin{aligned}\bar{y}_w &= \sum_{h=1}^H W_h \bar{y}_h = \sum_{h=1}^H \left(\frac{N_h}{N} \right) \bar{y}_h = \frac{1}{N} \sum_{h=1}^H N_h \frac{y_h}{n_h} = \frac{1}{N} \sum_{h=1}^H \frac{N_h}{n_h} \sum_{i=1}^{n_h} y_{hi} = \\ &= \frac{1}{N} \sum_{h=1}^H \sum_{i=1}^{n_h} \left(\frac{N_h}{n_h} \right) y_{hi} = \frac{1}{N} \sum_{h=1}^H \sum_{i=1}^{n_h} w_{hi} y_{hi} = \frac{\sum_{h=1}^H \sum_{i=1}^{n_h} w_{hi} y_{hi}}{\sum_{h=1}^H \sum_{i=1}^{n_h} w_{hi}}\end{aligned}$$

- Here,

$$\sum_{h=1}^H \sum_{i=1}^{n_h} w_{hi} = \sum_{h=1}^H \sum_{i=1}^{n_h} \left(\frac{N_h}{n_h} \right) = \sum_{h=1}^H n_h \left(\frac{N_h}{n_h} \right) = \sum_{h=1}^H N_h = N$$

Sampling variance

- Variances are combined across strata

$$\begin{aligned}\text{Var}(\bar{y}_w) &= \text{Var}\left(\sum_{h=1}^H W_h \bar{y}_h\right) \\ &= \text{Var}(W_1 \bar{y}_1 + W_2 \bar{y}_2 + \cdots + W_H \bar{y}_H) \\ &= \text{Var}(W_1 \bar{y}_1) + \text{Var}(W_2 \bar{y}_2) + \cdots + \text{Var}(W_H \bar{y}_H) \\ &= W_1^2 \text{Var}(\bar{y}_1) + W_2^2 \text{Var}(\bar{y}_2) + \cdots + W_H^2 \text{Var}(\bar{y}_H) \\ &= \sum_{h=1}^H W_h^2 \text{Var}(\bar{y}_h)\end{aligned}$$

- Estimate this sampling variance with $\text{var}(\bar{y}_w) = \sum_{h=1}^H W_h^2 \text{var}(\bar{y}_h)$

SRS within strata: Stratified Random Sampling

- Adding strata subscript, $\text{var}(\bar{y}_h) = \frac{1-f_h}{n_h} s_h^2$
- And combining across strata:

$$\text{var}(\bar{y}_w) = \sum_{h=1}^H W_h^2 \text{var}(\bar{y}_h) = \sum_{h=1}^H W_h^2 \frac{1-f_h}{n_h} s_h^2$$

- For proportions,

$$\text{var}(p_w) = \sum_{h=1}^H W_h^2 \left(\frac{1-f_h}{n_h-1} \right) p_h (1-p_h)$$

Confidence Intervals

- Confidence intervals:

$$\bar{y}_w \pm t_{1-\alpha/2, df} \sqrt{\text{var}(\bar{y}_w)} = \bar{y}_w \pm t_{1-\alpha/2, df} \sqrt{\sum_{h=1}^H W_h^2 \frac{1-f_h}{n_h} s_h^2}$$

$$p_w \pm t_{1-\alpha/2, df} \sqrt{\text{var}(p_w)} = p_w \pm t_{1-\alpha/2, df} \sqrt{\sum_{h=1}^H W_h^2 \left(\frac{1-f_h}{n_h-1} \right) p_h (1-p_h)}$$

$$\text{where } df = \sum_{h=1}^H (n_h - 1) = n - H$$

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	n_h	$\bar{y}_h$	s_h^2
1	< 50K sq. ft.	1,182	100	72.7	11,783.6
2	50 - 250K	1,846	200	336.1	205,657.8
3	250 - 4,000K	1,395	120	1,690.2	3,568,970.7
4	$\geq 4,000K$	895	80	11,659.0	183,039,437.2
Total		5,318	500		

$$\begin{aligned}\bar{y}_w &= \sum_{h=1}^H W_h \bar{y}_h = W_1 \bar{y}_1 + W_2 \bar{y}_2 + W_3 \bar{y}_3 + W_4 \bar{y}_4 = \\ &= \frac{1,182}{5,318} \times 72.7 + \frac{1,846}{5,318} \times 336.1 + \frac{1,395}{5,318} \times 1,690.2 + \frac{895}{5,318} \times 11,659.0 = 2,538.4\end{aligned}$$

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	n_h	$\bar{y}_h$	s_h^2
1	< 50K sq. ft.	1,182	100	72.7	11,783.6
2	50 - 250K	1,846	200	336.1	205,657.8
3	250 - 4,000K	1,395	120	1,690.2	3,568,970.7
4	$\geq 4,000K$	895	80	11,659.0	183,039,437.2
Total		5,318	500		

$$\begin{aligned} \text{var}(\bar{y}_w) &= \sum_{h=1}^H W_h^2 \text{var}(\bar{y}_h) = \sum_{h=1}^H W_h^2 \frac{1-f_h}{n_h} s_h^2 = \\ &= \left[\left(\frac{1,182}{5,318} \right)^2 \times \frac{1-\frac{100}{5,318}}{100} \times 11,783.6 \right] + \dots + \left[\left(\frac{895}{5,318} \right)^2 \times \frac{1-\frac{80}{5,318}}{80} \times 183,039,437.2 \right] = 60,998.0 \end{aligned}$$

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	n_h	$\bar{y}_h$	s_h^2
1	< 50K sq. ft.	1,182	100	72.7	11,783.6
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3	250 - 4,000K	1,395	120	1,690.2	3,568,970.7
4	$\geq 4,000K$	895	80	11,659.0	183,039,437.2
Total		5,318	500		

$$\bar{y}_w \pm t_{1-0.05/2, n-H} \sqrt{\text{var}(\bar{y}_w)} = 2,538.4 \pm 1.96 \sqrt{60,998.0} = \\ = [2,053.1; 3,023.6]$$

Unit 3: Stratification

Stratified Random Sampling

- A. Purpose of stratification
- B. Notation
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D. Precision & design effect

- What can we gain from this more complicated method of sample selection and estimation?
 - Better control, administrative convenience
 - Reduced variance – in some allocations
- How can we measure such potential gains in precision?
 - Compare sampling variance of the stratified estimate against its corresponding SRS sampling variance – Design effect
- Total variance in the population:

$$(N-1)S^2 = \sum_{h=1}^H \sum_{i=1}^{N_h} (Y_{hi} - \bar{Y})^2$$

Analysis of variance

- Expanding the total variance

$$\begin{aligned}\sum_{h=1}^H \sum_{i=1}^{N_h} (Y_{hi} - \bar{Y})^2 &= \sum_{h=1}^H \sum_{i=1}^{N_h} (Y_{hi} - \bar{Y}_h + \bar{Y}_h - \bar{Y})^2 \\&= \sum_{h=1}^H \sum_{i=1}^{N_h} \left[(Y_{hi} - \bar{Y}_h) + (\bar{Y}_h - \bar{Y}) \right]^2 \\&= \sum_{h=1}^H \sum_{i=1}^{N_h} \left[(Y_{hi} - \bar{Y}_h) \right]^2 + \sum_{h=1}^H \sum_{i=1}^{N_h} \left[(\bar{Y}_h - \bar{Y}) \right]^2 \\&\quad + 2 \sum_{h=1}^H \sum_{i=1}^{N_h} \left[(Y_{hi} - \bar{Y}_h) (\bar{Y}_h - \bar{Y}) \right]\end{aligned}$$

Components of variance

- Further simplification leads to...

$$\begin{aligned} & \sum_{h=1}^H \sum_{i=1}^{N_h} \left(\frac{N_h - 1}{N_h - 1} \right) (Y_{hi} - \bar{Y}_h)^2 + \sum_{h=1}^H N_h (\bar{Y}_h - \bar{Y})^2 \\ &= \sum_{h=1}^H (N_h - 1) S_h^2 + \sum_{h=1}^H N_h (\bar{Y}_h - \bar{Y})^2 \\ S^2 &= \sum_{h=1}^H \frac{(N_h - 1)}{(N - 1)} S_h^2 + \sum_{h=1}^H \left(\frac{N_h}{N - 1} \right) (\bar{Y}_h - \bar{Y})^2 \\ &\approx \sum_{h=1}^H W_h S_h^2 + \sum_{h=1}^H W_h (\bar{Y}_h - \bar{Y})^2 \\ &= \text{Within Variance} + \text{Between Variance} \end{aligned}$$

Designing stratified samples

- The sampling variance for the mean of a stratified sample is based on the within component only
 - Between stratum variance is ignored completely
- Decrease the sampling variance of the mean by making strata heterogeneous between and homogeneous within
- Use auxiliary X correlated with Y , since Y unknown
- Gains are not guaranteed
 - Depends on allocation of sample to strata - values of n_h

Design effect

- Gains (or losses) in precision can be measured through the design effect (*deff*):

$$deff = \frac{\text{var}(\bar{y}_w)}{\text{var}_{SRS}(\bar{y})} = \frac{\sum_{h=1}^H W_h^2 \left(\frac{1-f_h}{n_h} \right) s_h^2}{\left(\frac{1-f}{n} \right) s^2}$$

- Gain in precision if $deff < 1$
- Losses in precision $deff > 1$

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	n_h	$\bar{y}_h$	s_h^2
1	< 50K sq. ft.	1,182	100	72.7	11,783.6
2	50 - 250K	1,846	200	336.1	205,657.8
3	250 - 4,000K	1,395	120	1,690.2	3,568,970.7
4	$\geq 4,000K$	895	80	11,659.0	183,039,437.2
Total		5,318	500		

$$deff = \frac{\text{var}(\bar{y}_w)}{\text{var}_{SRS}(\bar{y})} = \frac{\sum_{h=1}^H W_h^2 \frac{1-f_h}{n_h} s_h^2}{\frac{1-f}{n} s^2} = \frac{60,988.0}{1 - \frac{500}{5,318} \times 49,038,472.8} = 0.6865$$

$$s^2 \approx \sum_{h=1}^H W_h s_h^2 + \sum_{h=1}^H W_h (\bar{y}_h - \bar{y})^2 = \left[\frac{1,182}{5,318} \times 11,783.6 + \dots \right] + \left[\frac{1,182}{5,318} \times (72.7 - 2,583.4)^2 + \dots \right] = 49,038,472.8$$

Unit 3: Stratification Allocations

- A. Proportionate
- B. Equal
- C. Neyman
- D. Optimum

Unit 3: Stratification Allocations

- A. Proportionate
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A. Proportionate

- Considering

$$\sum_{h=1}^H n_h = n \quad f_h = n_h / N_h \quad f = n / N$$

- If $f_h = f$ for all strata, *epsem* sampling results

- Hence,
$$\frac{n_h}{N_h} = \frac{n}{N} \Rightarrow n_h = n \times \frac{N_h}{N} = n \times W_h$$

- Self-weighting design: No weighting needed for the mean
- Sampling variance is simplified:

$$\text{var}(\bar{y}_w) = \left(\frac{1-f}{n} \right) \sum_{h=1}^H W_h s_h^2 = \left(\frac{1-f}{n} \right) s_w^2$$

Design effect

- Deff can be expressed as

$$deff = \frac{\text{var}(\bar{y}_w)}{\text{var}_{SRS}(\bar{y})} = \frac{\left(\frac{1-f}{n}\right) s_w^2}{\left(\frac{1-f}{n}\right) s^2} = \frac{s_w^2}{s^2}$$

- An alternate expression shows how gains can be obtained:

$$deff = 1 - \frac{\sum_{h=1}^H W_h (\bar{y}_h - \bar{y}_w)^2}{s^2}$$

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	W_h	n_h
1	< 50K sq. ft.	1,182	0.2223	111
2	50 - 250K	1,846	0.3471	174
3	250 - 4,000K	1,395	0.2623	131
4	$\geq 4,000K$	895	0.1683	84
Total		5,318		500

$$W_1 = \frac{N_1}{N} = \frac{1,182}{5,318} = 0.2223$$

$$n_1 = n \times \frac{N_1}{N} = 500 \times 0.2223 = 111.132 \approx 111$$

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	W_h	n_h	s_h^2
1	< 50K sq. ft.	1,182	0.2223	111	11,783.6
2	50 - 250K	1,846	0.3471	174	205,657.8
3	250 - 4,000K	1,395	0.2623	131	3,568,970.7
4	$\geq 4,000K$	895	0.1683	84	183,039,437.2
Total		5,318		500	

$$\text{var}(\bar{y}_w) = \left(\frac{1-f}{n} \right) s_w^2 = \left(\frac{1 - \frac{500}{5,318}}{500} \right) \times [0.2223 \times 11,783.6 + \dots] = 57,758.2$$

$$deff = \frac{\text{var}(\bar{y}_w)}{\text{var}_{SRS}(\bar{y})} = \frac{57,758.2}{\left(1 - \frac{500}{5,318} \right) \frac{49,038,472.8}{500}} = 0.65$$

Unit 3: Stratification Allocations

- A. Proportionate
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Unit 3: Stratification Allocations

- A. Proportionate
- B. Equal**
- C. Neyman
- D. Optimum

B. Equal

- Consider $n_h = \frac{n}{H}$ and $f_h = (n/H)/N_h$
- Not *epsem* unless strata are the same size
- Estimation requires the full weighted expressions
- *deff* may actually be greater than 1
- Why use it?
 - Suppose $S_h = S_{h'}$ for two different strata $h \neq h'$
 - Equal allocation minimizes the sampling variance $\text{var}(\bar{y}_h - \bar{y}_{h'})$

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	W_h	n_h	s_h^2
1	< 50K sq. ft.	1,182	0.2223	125	11,783.6
2	50 - 250K	1,846	0.3471	125	205,657.8
3	250 - 4,000K	1,395	0.2623	125	3,568,970.7
4	$\geq 4,000K$	895	0.1683	125	183,039,437.2
Total		5,318		500	

$$\begin{aligned} \text{var}(\bar{y}_w) &= \sum_{h=1}^H W_h^2 \text{var}(\bar{y}_h) = \sum_{h=1}^H W_h^2 \frac{1-f_h}{n_h} s_h^2 \\ &= \left[0.2223^2 \times \frac{1-\frac{125}{1,182}}{125} \times 11,783.6 \right] + \dots + \left[0.1683^2 \times \frac{1-\frac{125}{895}}{125} \times 183,039,437.2 \right] = 49,038,472.8 \end{aligned}$$

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	W_h	n_h	s_h^2
1	< 50K sq. ft.	1,182	0.2223	125	11,783.6
2	50 - 250K	1,846	0.3471	125	205,657.8
3	250 - 4,000K	1,395	0.2623	125	3,568,970.7
4	$\geq 4,000K$	895	0.1683	125	183,039,437.2
Total		5,318		500	

$$deff = \frac{\text{var}(\bar{y}_w)}{\text{var}_{SRS}(\bar{y})} = \frac{49,038,472.8}{\left(1 - \frac{500}{5,318}\right) \frac{49,038,472.8}{500}} = 0.4238$$

Unit 3: Stratification Allocations

- A. Proportionate
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Unit 3: Stratification Allocations

- A. Proportionate
- B. Equal
- C. Neyman
- D. Optimum

C. Neyman

- Consider $n_h = n \times W_h S_h / \sum_{h=1}^H W_h S_h$
- Smallest variance for $\bar{y}_w$
- Gains in precision greater than proportionate
- Need estimates of S_h
 - In practice, use reasonable estimates
 - Large gains require variation among S_h
 - Large gains unlikely for proportions
 - Values specific to Y
 - For multipurpose surveys, allocations will vary as S_h vary across characteristics

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	W_h	s_h^2	n_h
1	< 50K sq. ft.	1,182	0.2223	11,783.6	4
2	50 - 250K	1,846	0.3471	205,657.8	27
3	250 - 4,000K	1,395	0.2623	3,568,970.7	84
4	$\geq 4,000K$	895	0.1683	183,039,437.2	385
Total		5,318			500

$$n_1 = n \times \frac{W_1 s_1}{\sum_{h=1}^4 W_h s_h} = 500 \times \frac{0.2223 \times \sqrt{11,783.6}}{0.2223 \times \sqrt{11,783.6} + \dots + 0.1683 \times \sqrt{183,039,437.2}} = 4.084 \approx 4$$

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	W_h	s_h^2	n_h
1	< 50K sq. ft.	1,182	0.2223	11,783.6	4
2	50 - 250K	1,846	0.3471	205,657.8	27
3	250 - 4,000K	1,395	0.2623	3,568,970.7	84
4	$\geq 4,000K$	895	0.1683	183,039,437.2	385
Total		5,318			500

$$\text{var}(\bar{y}_w) = \sum_{h=1}^H W_h^2 \text{var}(\bar{y}_h) = \sum_{h=1}^H W_h^2 \frac{1-f_h}{n_h} s_h^2$$

$$= \left[0.2223^2 \times \frac{1 - \frac{4}{1,182}}{4} \times 11,783.6 \right] + \dots + \left[0.1683^2 \times \frac{1 - \frac{385}{895}}{385} \times 183,039,437.2 \right] = 11,470.2$$

Example: Stratified sample data, $n = 500$

Stratum h	Group	N_h	W_h	s_h^2	n_h
1	< 50K sq. ft.	1,182	0.2223	11,783.6	4
2	50 - 250K	1,846	0.3471	205,657.8	27
3	250 - 4,000K	1,395	0.2623	3,568,970.7	84
4	$\geq 4,000K$	895	0.1683	183,039,437.2	385
Total		5,318			500

$$deff = \frac{\text{var}(\bar{y}_w)}{\text{var}_{SRS}(\bar{y})} = \frac{11,470.2}{\left(1 - \frac{500}{5,318}\right) \frac{49,038,472.8}{500}} = 0.1291$$

Unit 3: Stratification Allocations

- A. Proportionate
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Unit 3: Stratification Allocations

- A. Proportionate
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D. Optimum

- Consider total fixed cost $C = \sum_{h=1}^H n_h J_h$
- Objective:
 - Minimize sampling variance for $\bar{y}_w$ subjected to total fixed cost C , **OR**
 - Minimize cost subjected to a specified sampling variance for $\bar{y}_w$
- Allocate $n_h = k W_h S_h / \sqrt{J_h}$
 - k is a constant of proportionality
- Neyman allocation is a special case where costs are the same across strata

Optimum allocation - Steps

- For a total fixed cost C :
 - Compute preliminary allocation with $n_h^* = W_h S_h / \sqrt{J_h}$
 - Calculate preliminary cost under that allocation: $C^* = \sum_{h=1}^H n_h^* J_h$
 - Compute constant of proportionality as $k = C / C^*$
 - Adjust preliminary allocation $n_h = k \times n_h^*$
- Similar process for fixed sampling variance

Example: $C = \$150,000$ for Optimum allocation

Stratum h	Group	W_h	S_h^2	J_h	n_h^*	n_h
1	< 50K sq. ft.	0.2223	11,783.6	100	2.413	7
2	50 - 250K	0.3471	205,657.8	100	15.742	43
3	250 - 4,000K	0.2623	3,568,970.7	200	35.041	97
4	$\geq 4,000K$	0.1683	183,039,437.2	400	113.846	314
Total						461

$$n_1^* = W_1 s_1 / \sqrt{J_1} = 0.2223 \times \sqrt{11,783.6} / \sqrt{100} = 2.413$$

$$C^* = \sum_{h=1}^H n_h^* J_h = 2.413 \times 100 + 15.742 \times 100 + 35.041 \times 200 + 113.846 \times 400 = 54,362.08$$

$$k = C / C^* = 150,000 / 54,362.08 = 2.759$$

$$n_1 = k \times n_1^* = 2.759 \times 2.413 = 6.657 \approx 7$$

Example: $C = \$150,000$ for Optimum allocation

Stratum h	Group	W_h	s_h^2	J_h	n_h
1	< 50K sq. ft.	0.2223	11,783.6	100	7
2	50 - 250K	0.3471	205,657.8	100	43
3	250 - 4,000K	0.2623	3,568,970.7	200	97
4	$\geq 4,000K$	0.1683	183,039,437.2	400	314
Total					461

$$\text{var}(\bar{y}_w) = \sum_{h=1}^H W_h^2 \text{var}(\bar{y}_h) = \sum_{h=1}^H W_h^2 \frac{1-f_h}{n_h} s_h^2$$

$$= \left[0.2223^2 \times \frac{1-\frac{7}{461}}{7} \times 11,783.6 \right] + \dots + \left[0.1683^2 \times \frac{1-\frac{314}{461}}{314} \times 183,039,437.2 \right] = 13,719.3$$

Example: $C = \$150,000$ for Optimum allocation

Stratum h	Group	W_h	S_h^2	J_h	n_h
1	< 50K sq. ft.	0.2223	11,783.6	100	7
2	50 - 250K	0.3471	205,657.8	100	43
3	250 - 4,000K	0.2623	3,568,970.7	200	97
4	$\geq 4,000K$	0.1683	183,039,437.2	400	314
Total					461

$$deff = \frac{\text{var}(\bar{y}_w)}{\text{var}_{SRS}(\bar{y})} = \frac{13,719.3}{\left(1 - \frac{461}{5,318}\right) \frac{49,038,472.8}{461}} = 0.1412$$

Unit 3: Stratification

Stratification Topics

- A. Number of strata
- B. Paired selection
- C. Collapsed strata
- D. Poststratification
- E. Sub-classes
- F. Step-by-step selection

Unit 3: Stratification

Stratification Topics

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A. Number of strata

- Stratification requires discrete categories
 - Stratifying variables may be discrete
 - Continuous stratifying variables divided into categories
- How many categories to capture gains possible?
 - Generally 3-6 strata adequate
 - When more than one stratifier, "coarser" cuts on more variables preferred to "finer" cuts
 - "Deepest" stratification for $n_h = 2$ (or $H = n/2$)
 - "Even deeper" stratification: 1 per stratum

Unit 3: Stratification

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Unit 3: Stratification

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B. Paired selection

- Paired selections $n_h = 2$ useful in practice
 - “Deepest” stratification possible that allows sampling variances to be estimated without assumptions
 - Paired selection is *epsem* when: $N_h = N/H$
 - Attraction of paired selection is the simplification in variance estimation
 - When the design is *epsem*, proportionately allocated

Paired selection estimation

$$\bar{y} = \frac{\sum_{h=1}^H \sum_{i=1}^{n_h} y_{hi}}{n} = \frac{\sum_{h=1}^H (y_{h1} + y_{h2})}{n}$$

$$\begin{aligned} \text{var}(\bar{y}) &= \left(\frac{1-f}{n} \right) \sum_{h=1}^H W_h s_h^2 \quad \left(\text{where } W_h = \frac{N_h}{N} = \frac{n_h}{n} = \frac{2}{n} \right) \\ &= \left(\frac{1-f}{n} \right) \sum_{h=1}^H \frac{2}{n} \left[\left(y_{h1} - \frac{y_{h1} + y_{h2}}{2} \right)^2 + \left(y_{h2} - \frac{y_{h1} + y_{h2}}{2} \right)^2 \right] \\ &= 2 \left(\frac{1-f}{n^2} \right) \sum_{h=1}^H \left[\left(\frac{y_{h1} - y_{h2}}{2} \right)^2 + \left(\frac{y_{h2} - y_{h1}}{2} \right)^2 \right] \\ &= \left(\frac{1-f}{n^2} \right) \sum_{h=1}^H (y_{h1} - y_{h2})^2 \end{aligned}$$

Paired selection features

- For element sampling, the symmetry of the selection and variance estimation disrupted by
 - Blanks in the list, non-responding elements
 - Analysis of subclasses
- Remedy: collapse strata
 - Increases the number of selections within "collapsed" or "pseudo" strata
 - Sampling variances are estimated as though the stratum boundary(ies) do not exist
 - “Deeper” stratification, which can increase the precision of estimates, is ignored
 - Sampling variance is over-estimated.

Unit 3: Stratification

Stratification Topics

- A. Number of strata
- B. Paired selection
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- E. Sub-classes
- F. Step-by-step selection

Unit 3: Stratification

Stratification Topics

- A. Number of strata
- B. Paired selection
- C. Collapsed strata**
- D. Poststratification
- E. Sub-classes
- F. Step-by-step selection

C. Collapsed strata

Stratum	Elements	Collapsed stratum	Elements
1	y_{11}, y_{12}	1	$y_{11}, y_{12}, y_{21}, y_{22}$
2	y_{21}, y_{22}		
3	y_{31}, y_{32}	2	$y_{31}, y_{32}, y_{41}, y_{42}$
4	y_{41}, y_{42}		

Collapsed strata properties

- Used to simplify more complicated designs
- Reduced number of strata in estimation
- Increased degrees of freedom ($n - H$)
- Variance estimate based on collapsed strata biased for variance for original strata
 - Over-estimates variance by factor proportionate to between stratum difference

Unit 3: Stratification

Stratification Topics

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Unit 3: Stratification

Stratification Topics

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- E. Sub-classes
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D. Poststratification

- Poststratification: variables to be used to create strata are not available at the time of selection
 - Stratify after selection using variables collected during the survey
 - Gains in precision are possible, with suitable modification to variance estimation
 - Population control adjustment
 - Poststratification requires
 - Poststrata known for each element
 - Poststratum weights W_g for each poststratum

Poststratified estimator

- Sort the sample into poststrata
- Compute poststratum means
- Compute poststratified mean as $\bar{y}_{ps} = \sum_{g=1}^G W_g \bar{y}_g$
- For the variance,

$$\text{Var}(\bar{y}_{ps}) \approx \left(\frac{1-f}{n} \right) \left[\sum_{g=1}^G W_g S_g^2 + \sum_{g=1}^G W_g (1-W_g) \frac{S_g^2}{n_g} \right]$$

Poststratification illustration

- Sample $n=2,000$ SRS from $N=20,000$ libraries
 - $g=1$: ordered last year and prior years
 - $g=2$: ordered last year only
 - $g=3$: never ordered

g	W_g	n_g	n_g/n	$\bar{y}_g$	s_g^2
1	0.20	410	0.205	1250	3.2×10^6
2	0.25	493	0.247	750	3.0×10^6
3	<u>0.55</u>	<u>1097</u>	<u>0.548</u>	<u>140</u>	<u>1.5×10^6</u>
		2000	1.000		$(s^2 = 2.4 \times 10^6)$

Estimation

- The mean and variance are respectively

$$\bar{y}_{ps} = (0.2)(1250) + (0.25)(750) + (0.55)(140) = 514.5$$

$$\text{var}(\bar{y}_{ps}) = \left(\frac{1-0.1}{2000} \right) \left[(0.2)(3.2 \times 10^6) + (0.2)(1-0.2) \frac{3.2 \times 10^6}{410} + \dots \right] = 998.0$$

- Under proportionate allocation

$$\text{var}(\bar{y}) = \left(\frac{1-f}{n} \right) \sum_{g=1}^G W_g s_g^2 = 996.8$$

- Under SRS

$$\text{var}_{SRS}(\bar{y}) = \left(\frac{1-f}{n} \right) s^2 = 1086.4$$

Poststratification gains in precision

- The design effects are ...

$$deff_1 = \frac{\text{var}(\bar{y}_{ps})}{\text{var}_{SRS}(\bar{y})} = 0.9575 \quad deff_2 = \frac{\text{var}(\bar{y})}{\text{var}_{SRS}(\bar{y})} = 0.9563$$

- W_g may be approximate
- Use of the W_g and poststratified estimators are a bias reduction technique
- Multipurpose poststratification limits choice of W_g
- Variance estimate: Taylor series approximation

Unit 3: Stratification

Stratification Topics

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Unit 3: Stratification

Stratification Topics

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- F. Step-by-step selection

E. Sub-classes

- Stratification gains diminish for subclasses
- Let M = total number of subclass elements in the population
- M_h denotes the number for stratum h
- m = subclass sample size (not "controlled")
- m_h is the size for stratum h
- The relative size of the subclass is given by

$$\bar{M} = \frac{M}{N} \text{ and } \bar{m} = \frac{m}{n}$$

Stratified sub-class estimator

- Consider the subclass mean for stratified random sampling
- The individual M_h will not be known
- Estimate M_h from the sample as : $\hat{M}_h = F_h m_h = \frac{N_h}{n_h} m_h$

Losses in gains in precision

- Subclass mean "loses" gains in precision
- For large subclasses, $1 - \bar{m}_h$ small, gains retained
- For small subclasses, $1 - \bar{m}_h = 1$, gains lost
- Resulting estimate no more precise than an SRS of the subclass
- Loss is approximately

$$\left(1 - m'_h\right) \sum_{h=1}^H \left(w'_h\right)^2 \left(\bar{y}_h - \bar{y}_w\right)^2$$

Unit 3: Stratification

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Unit 3: Stratification

Stratification Topics

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- F. Step-by-step selection

F. Step-by-step

- The "steps" in stratified sampling are as follows:
 - Identify stratifying variables correlated with the measure(s) of interest
 - Choose "cuts" on the stratifying variables
 - Divide the population into strata
 - S_h , $deff$, and S^2 are known, at least approximately, and that fpc 's within strata can be ignored
 - Compute an SRS sample size $n = n_{SRS} \times deff$

Finishing the design

- Determine an allocation for the desired n
 - Proportionate
 - Neyman or optimum
 - Equal
- Adjust SRS n based on expected $deff$ & allocate
- Select sample and compute estimates taking the stratified sample selection into account
 - Mean
 - Variance

Unit 4: Systematic Selection

- A. Principles
- B. Fractional Intervals
- C. Implicit Stratification
- D. Variance Estimation

Unit 4: Systematic Selection

- A. Principles
- B. Fractional Intervals
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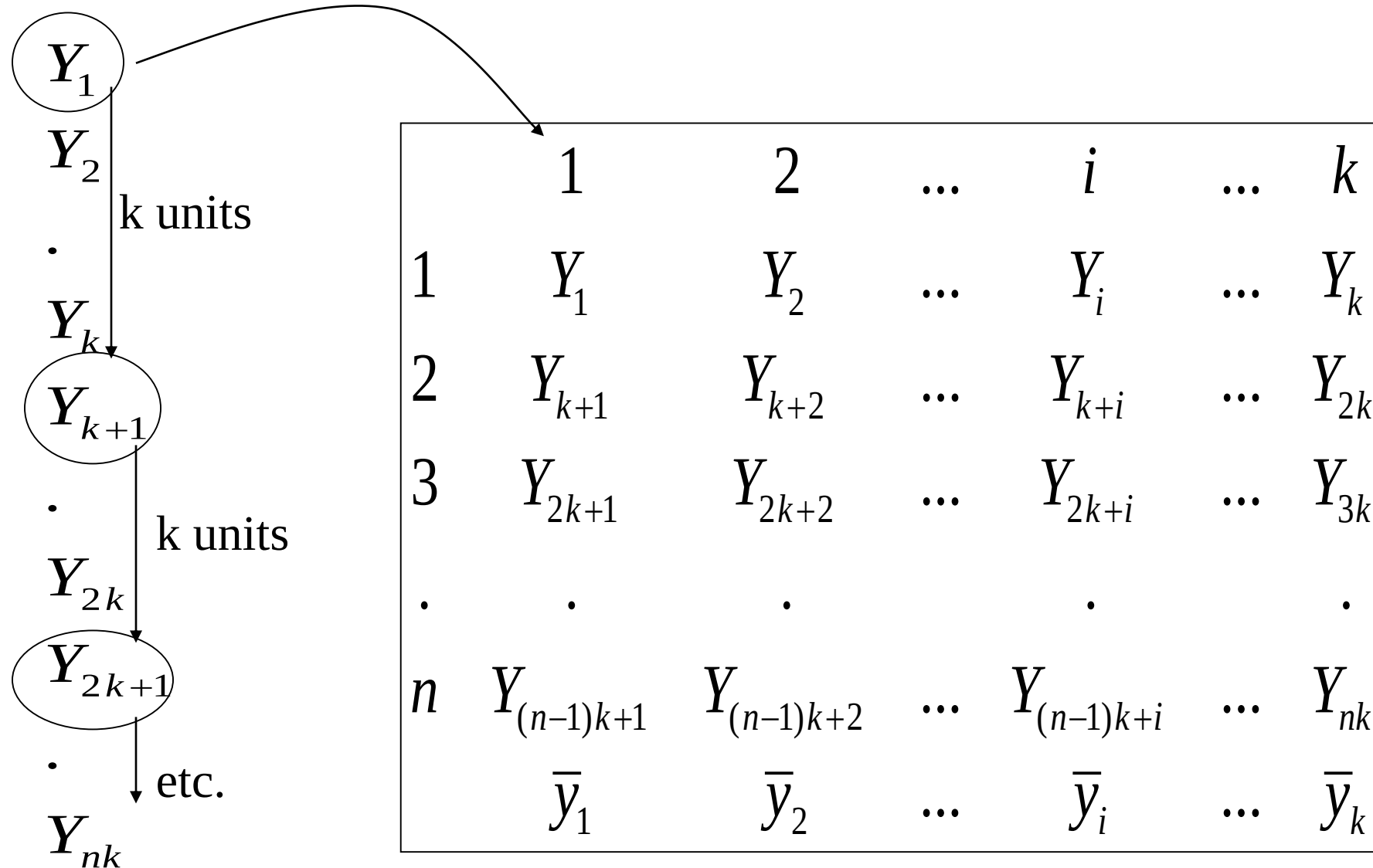
A. Principles

- SRS is difficult to implement and check
 - Without replacement selection
- Systematic sampling is clerically easy to do
 - There are problems in implementation
- Consider systematic sampling from a numbered list with N population elements, n sample elements, and $f = n/N$

Sample selection

- The sampling interval is $I = \frac{N}{n} = \frac{1}{f} = F = k$
- Selection starts with choosing a random start RS from 1 to k
 - Select population elements RS, RS+ k , RS+2 k , RS+3 k , . . . , RS+(n -1) k
- This selection process conceptually arrange population values as follows:

Conceptual framework

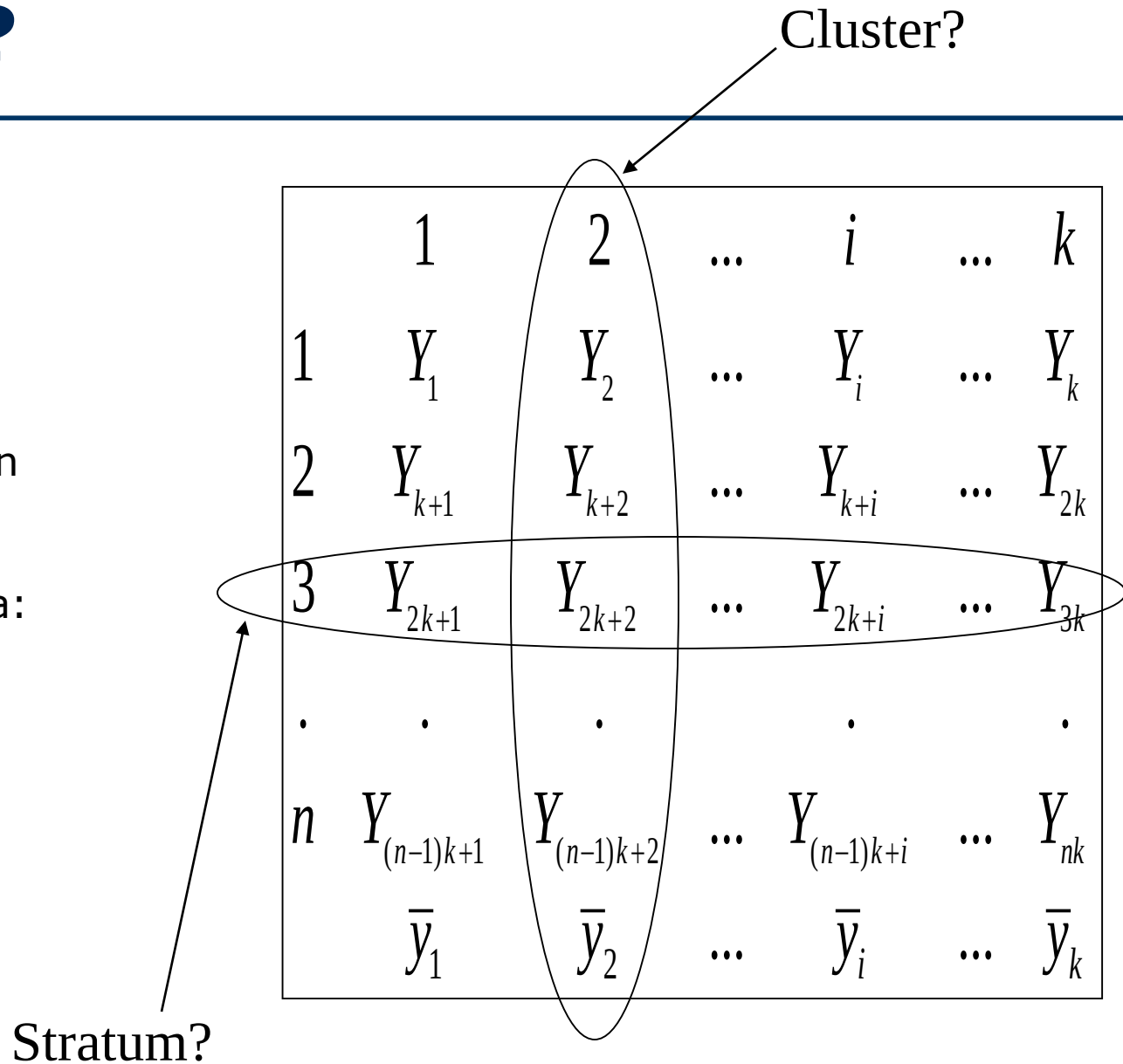


Illustration

	1	2	3	4	5	6
1	1	2	3	4	5	6
2	7	8	9	10	11	12
3	13	14	15	16	17	18
4	19	20	21	22	23	24
5	25	26	27	28	29	30
6	31	32	33	34	35	36
7	37	38	39	40	41	42

Clusters or strata?

- Order of the list achieves stratification
 - Combination of list order and systematic selection
 - Group similar elements together in list
 - Zones in selection similar to strata: contain elements similar to one another
 - Stratified proportionate allocation
- Systematic sampling is not a measurable design:
 - Stratified sample with $n_h=1$
 - Cluster sample with $a = 1$



Unit 4: Systematic Selection

- A. Principles
- B. Fractional Intervals
- C. Implicit Stratification
- D. Variance Estimation

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B. Fractional intervals

- Interval $k = N/n$ may not be an integer
 - Suppose that $N = 23$ and $n = 5$, so that $k = 23/5 = 4.6$
- If $k = 5$ is applied, $n = 4$ or 5 will be random

	1	2	3	4	5
1	1	2	3	4	5
2	6	7	8	9	10
3	11	12	13	14	15
4	16	17	18	19	20
5	21	22	23		

Solutions for fractional intervals (1)

- What to do?
- Adjust N or n to make k an integer.
- If the achieved sample size is $n+1$, delete a case at random from the sample
 - Unfortunately, this is not *epsem*
- Instead eliminate $N-nk$ listings at random, and then sample systematically
$$f = \left(1 - \frac{N - nk}{N}\right) \times \frac{n}{nk} = \frac{n}{N}$$
- Duplicating $n(k+1)-N$ elements at random also works

Solutions for fractional intervals (2)

- Consider the list to be circular
 - Select a RS from 1 to $N=23$ and continue selection until $n=5$
 - Example: RS=**17**
 - Add integer closest to k ($k=4.6$, so use 5)
 - Select $17+5=\mathbf{22}$
 - $22+5=27$, but $27>23$
 - Go to the beginning of the list and select **4**
 - $4+5=\mathbf{9}$, $9+5=\mathbf{14}$

Solutions for fractional intervals (3)

- Use the fractional interval
 - Choose a random start from 01 to 46 (implied one decimal place)
 - Truncate the decimal to make the selection
 - Add the interval 46 to the random start
 - Truncate the decimal . . . Continue to $n = 5$
 - If the random start is 01-09, the first selection will be obtained after the interval is added
 - Example: RS=35:
 - 35, $35+46=81$, $81+46=127$, $127+46=173$, $173+46=219$
 - Choose 3, 8, 12, 17, 21

Fractional interval summary

RS	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61
	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107
	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153
	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199
	231	232	233	234	235	236	237	238	239						

... ..

RS	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46
	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92
	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138
	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184
	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230

Fractional interval probabilities

Element	Random start	No. of RS's	f
1	10-19	10	1/4.6
2	20-29	10	1/4.6
3	30-39	10	1/4.6
4	40-46,01-03	10	1/4.6
5	04-13	10	1/4.6
...	...	...	...
23	01-09,46	10	1/4.6

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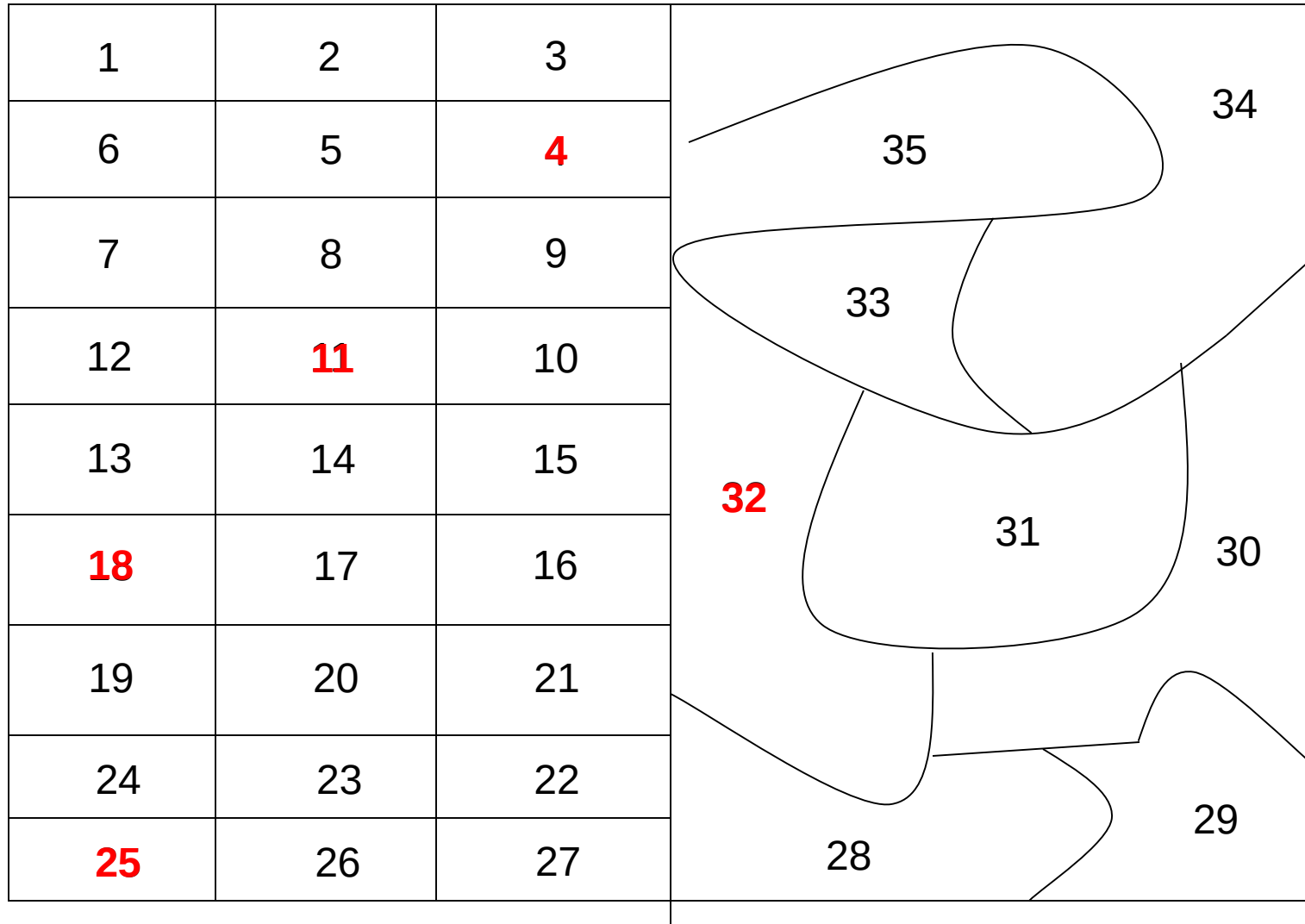
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C. Implicit stratification

- List order combined with systematic selection can improve the efficiency (in terms of variance) of systematic sample designs
 - Arrange the list order in advance
 - Determines which samples selected
 - Random order: SRS
 - Stratified order
 - Serpentine order

City blocks



$$n = 5$$
$$k = 35/5 = 7$$
$$RS = 4$$

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D. Variance estimation

- Estimation of the sample mean: $\bar{y} = \frac{\sum y_i}{n}$
- Sampling variance cannot be estimated using only survey data
 - Only a single random start used
 - Two approaches to dealing with the problem
 - Use additional random starts
 - Model the variance
 - Use c random starts and $\bar{y} = \frac{\sum \sum y_{ji}}{cn} = \frac{1}{c} \sum_{\gamma} \bar{y}_{\gamma}$

$$\text{var}(\bar{y}) = \frac{1}{c} \sum_{\gamma} (\bar{y}_{\gamma} - \bar{y})^2$$

Model assisted variance estimation

- Model the population (sample selection process):
- SRS model -- elements in the list are ordered at random.

$$\text{var}(\bar{y}) = (1 - f) \frac{s^2}{n}$$

Stratified random model

- Assumes homogeneity and random ordering within zones
- Proportionately allocated selection with $n_h = 1$ selected per zone
- Collapse neighboring zones to create "pseudo strata" that have more than one selection

$$\text{var}(\bar{y}) = \frac{1-f}{n} \sum W_h s_h^2$$

Paired selection model

- Special case of stratified random model which pairs successive zones
- Simple variance estimation expression
- Simplicity can be disrupted by nonresponse or estimation for a subclass

$$\text{var}(\bar{y}) = \frac{1-f}{n^2} \sum_{h=1}^{n/2} (y_{h1} - y_{h2})^2$$

Successive differences model

- Correlation between the i^{th} and $(i+1)^{\text{th}}$ element larger than the correlation between the i^{th} and $(i+2)^{\text{th}}$ or the i^{th} and $(i+3)^{\text{th}}$, etc.
- $(n-1)$ differences each weighted by $1/2$ (except the first and last)
 - Not unbiased variance estimate, but consistent
 - Bias is small for large n .

$$\text{var}(\bar{y}) = \frac{1-f}{2n(n-1)} \sum_{g=1}^{n-1} (y_g - y_{g+1})^2$$

Variance estimation example

- Take a systematic sample of $n=6$ from the first 60 blocks of the 270 blocks in Kish (p. 625)
 - How many systematic samples of $n=6$ are possible from the first 60 blocks?
 - What is the interval?
- Random start = 3
- What are the 6 selected blocks?

Selected blocks

RS+k	Block	# Rental	# HUs	g
4	242	79	90	1
14	279	84	89	2
24	289	36	45	3
34	378	19	29	4
44	389	21	32	5
54	399	12	18	6

Calculating means

- *Epsem* sample

$$\bar{y}_{\#rental} = \frac{\sum_{i=1}^n y_i}{n}$$

$$= (79 + 84 + 36 + 19 + 21 + 12) / 6 = 41.83$$

Calculating variances (1)

- SRS model

$$\begin{aligned}\text{var}(\bar{y}) &= (1 - f) \frac{s^2}{n} = \left(1 - \frac{6}{60}\right) \left(\frac{1}{6}\right) \frac{(15,539 - 6 * 41.83^2)}{6 - 1} \\ &= (0.90)(0.1667)(1,007.77) = 151.17\end{aligned}$$

Calculating variances (2)

- Paired selection model, even # elements

$$\begin{aligned}\text{var}(\bar{y}) &= \frac{(1-f)}{n^2} \sum_h^{n/2} (y_{ha} - y_{hb})^2 \\ &= \left(1 - \frac{6}{60}\right) \left(\frac{1}{6^2}\right) [(79-84)^2 + (36-19)^2 + (21-12)^2] \\ &= (.9)(0.0278)(25 + 289 + 81) = 9.875\end{aligned}$$

Calculating variances (3)

- Successive differences model

$$\begin{aligned}\text{var}(\bar{y}) &= \frac{1-f}{2n(n-1)} \sum_g^{n-1} (y_g - y_{g+1})^2 \\ &= \frac{1 - \cancel{6}/60}{2 * 6(6-1)} [(79-84)^2 + (84-36)^2 + (36-19)^2 \\ &\quad + (19-21)^2 + (21-12)^2] \\ &= \frac{.9}{60} [25 + 2,304 + 289 + 4 + 81] = 40.55\end{aligned}$$

Choosing among models

- Criteria for selecting an appropriate model include
 - List order
 - random (or "effectively random")
 - ordered deliberately to achieve stratification
 - Degrees of freedom
 - SRS: $n-1$
 - Successive differences: between $n/2$ & $n-1$
 - Use the more conservative $n/2$ in CIs
 - Stratified random: $n-H$
 - Paired differences: $n/2$

Bias considerations

- Variance estimates may be biased
 - Model fails to "capture" stratification represented by systematic selection
 - SRS: no stratification effects are accounted for
 - Stratified selection: captures stratification
 - Collapses across stratum boundaries
 - Paired selection: two per stratum
 - Successive differences: deepest stratification possible

Unit 5: Clusters

Cluster Sampling

- A. Clusters and survey costs
- B. Variance of the mean
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A. Cluster and survey costs

- Populations often distributed geographically
 - Cannot afford to visit n units drawn randomly from the entire area
 - Cannot afford to create an element frame
- Cluster selections are used to reduce costs
 - Select clusters and list elements only for selected clusters
- Clusters naturally occurring units:
 - Seldom equal size

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B. Variance of the mean

Target Population
Elements:

101 Main St.

104 Main St.

107 Main St.

112 Main St.

115 Main St.

122 Main St.

129 Main St.

132 Main St.

201 Main St.

206 Main St.

.

.

.

$$\bar{Y} = \sum_{i=1}^N Y_i / N$$

$$S^2 = \sum_{i=1}^N (Y_i - \bar{Y})^2 / (N - 1)$$

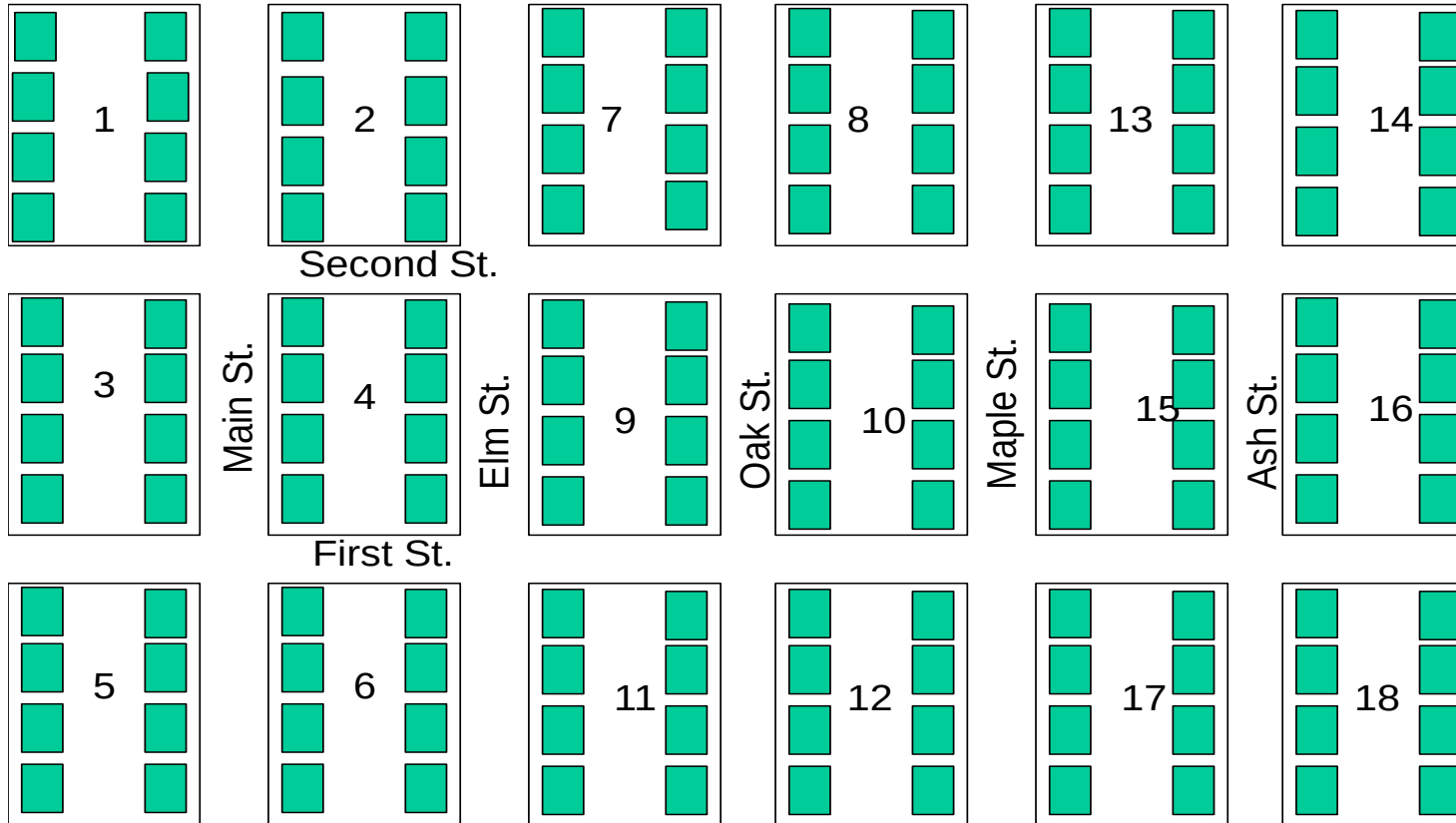
Notation

- A = the number of clusters in the population
- a = the number of clusters in the sample
- B = the number of elements in each cluster
- $\frac{a}{A}$ is the rate at which clusters are selected
- If we select all elements within a cluster, the rate is $\frac{B}{B} = 1$
- $\frac{a}{A} \times \frac{B}{B} = \frac{n}{N}$

Clustered population

$$\bar{Y}_\alpha = \sum_{\beta=1}^B Y_{\alpha\beta} / B$$

$$\bar{Y} = \sum_{i=1}^N Y_i / N$$



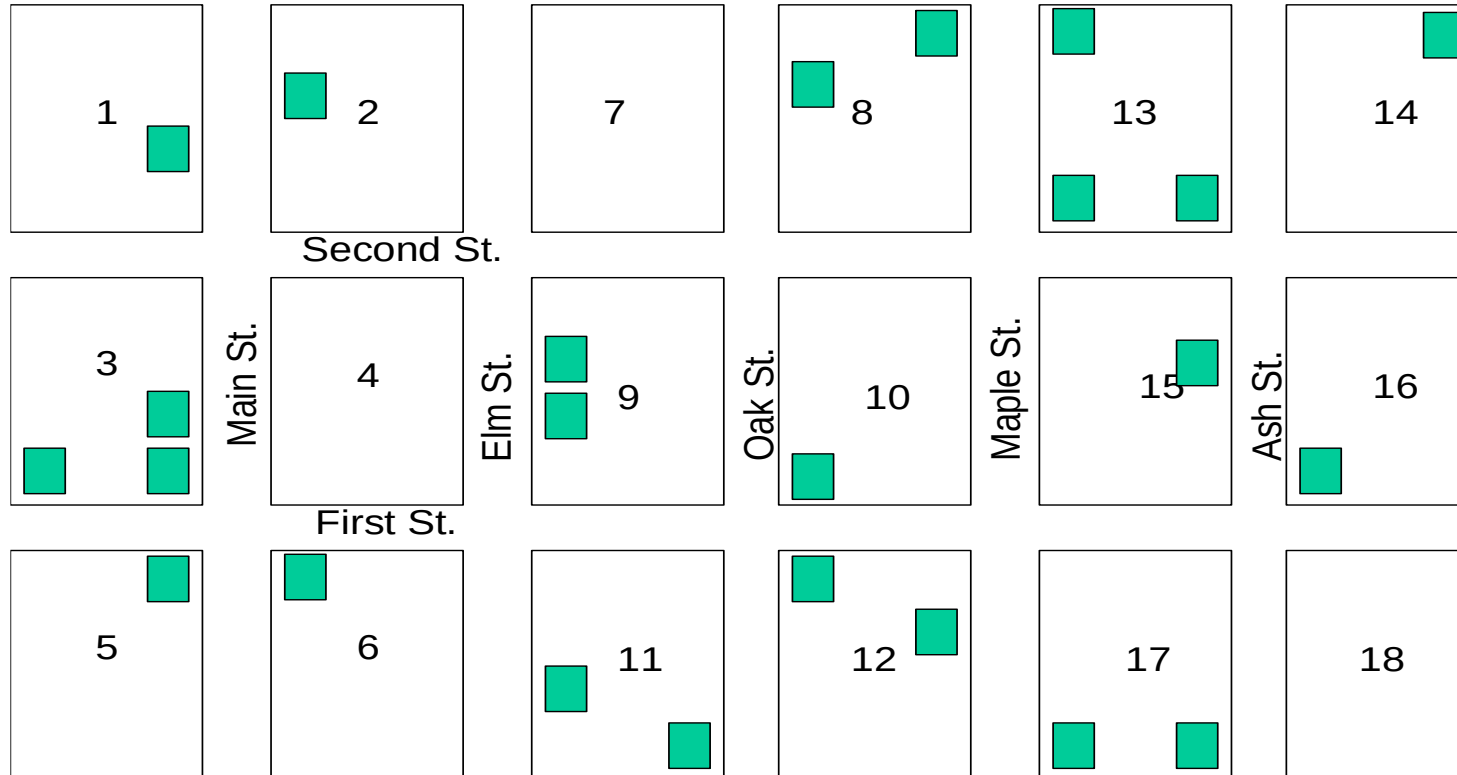
$$S_\alpha^2 = \sum_{\beta=1}^B (Y_{\alpha\beta} - \bar{Y}_\alpha)^2 / (B-1)$$

$$S^2 = \sum_{i=1}^N (Y_i - \bar{Y})^2 / (N-1)$$

Element sample

$$\bar{y} = \sum_{i=1}^n y_i / n$$

$$\bar{Y} = \sum_{i=1}^N Y_i / N$$



$$s^2 = \sum_{i=1}^n (y_i - \bar{y})^2 / (n-1)$$

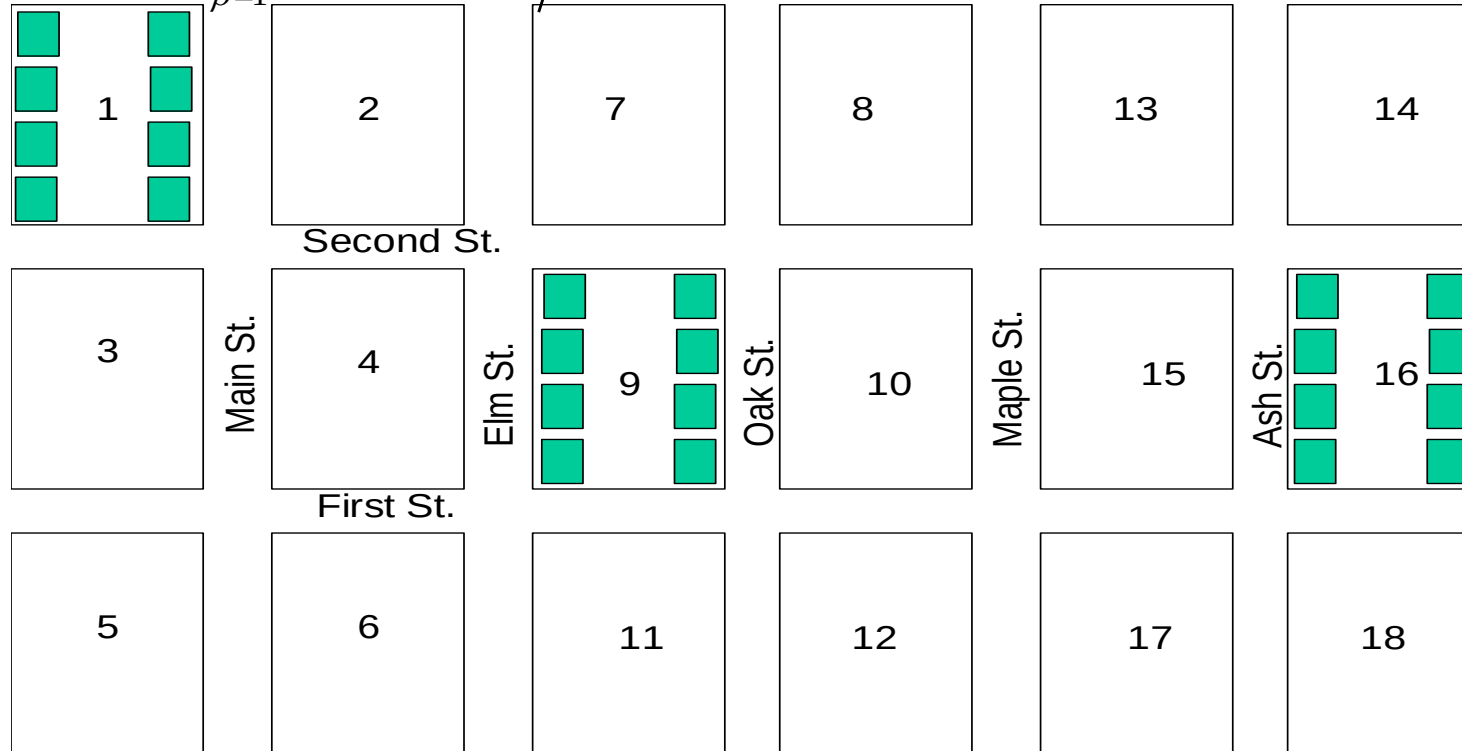
$$\text{Var}(\bar{y}) = \frac{(1-f)}{n} S^2$$

$$S^2 = \sum_{i=1}^N (Y_i - \bar{Y})^2 / (N-1)$$

Cluster sample

$$\bar{y} = \sum_{\alpha=1}^a \sum_{\beta=1}^B y_{\alpha\beta} / aB \quad s_{\alpha}^2 = \sum_{\beta=1}^B (y_{\alpha\beta} - \bar{y}_{\alpha})^2 / (B-1)$$

$$\bar{Y} = \sum_{i=1}^N Y_i / N$$



$$s_a^2 = \sum_{\alpha=1}^a (\bar{y}_{\alpha} - \bar{y})^2 / (a-1)$$

$$Var(\bar{y}) = \frac{(1-f)}{a} S_a^2$$

$$S^2 = \sum_{i=1}^N (Y_i - \bar{Y})^2 / (N-1)$$

Computing formulas

- Here we estimate

$$\text{var}(\bar{y}) = \frac{1-f}{a} s_a^2 = \frac{1-f}{a} \frac{\sum_{\alpha=1}^a (\bar{y}_{\alpha} - \bar{y})^2}{a-1}$$

- But there is an algebraic expansion...

$$\text{var}(\bar{y}) = \frac{1-f}{a} \frac{\left[\sum_{\alpha=1}^a \bar{y}_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a \bar{y}_{\alpha} \right)^2}{a} \right]}{a-1}$$

Computing formulas

- And that algebraic expansion is converted into one involving cluster totals:

$$\begin{aligned}\text{var}(\bar{y}) &= \frac{1-f}{a} \left[\frac{\sum_{\alpha=1}^a \bar{y}_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a \bar{y}_{\alpha}\right)^2}{a}}{a-1} \right] = \frac{1-f}{a(a-1)} \left[\sum_{\alpha=1}^a \left(\frac{y_{\alpha}}{B}\right)^2 - \frac{\left(\sum_{\alpha=1}^a \frac{y_{\alpha}}{B}\right)^2}{a} \right] \\ &= \frac{1-f}{aB^2(a-1)} \left[\sum_{\alpha=1}^a y_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a y_{\alpha}\right)^2}{a} \right]\end{aligned}$$

Computing formulas

- So, for taking complete clusters (simple cluster sampling),

$$\begin{aligned}\text{var}(\bar{y}) &= \frac{1-f}{a} s_a^2 \\ &= \frac{1-f}{a} \frac{\sum_{\alpha=1}^a (\bar{y}_{\alpha} - \bar{y})^2}{a-1} \\ &= \frac{1-f}{aB^2(a-1)} \left[\sum_{\alpha=1}^a y_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a y_{\alpha} \right)^2}{a} \right]\end{aligned}$$

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C. Design effect

- There is no particular reason $\text{var}(\bar{y}) \neq \text{var}_{SRS}(\bar{y})$
- For cluster sampling:

$$deff = \frac{\text{Var}(\bar{y})}{\text{Var}_{srs}(\bar{y})} = \frac{\frac{1-f}{a} S_a^2}{\frac{1-f}{n} S^2} = \frac{S_a^2 / a}{S^2 / n} > 1.0$$

Design effect: Alternate form

$$\begin{aligned} \text{Var}(\bar{y}) &= \text{Var}\left(\frac{1}{n} \sum_{\alpha} \sum_{\beta} y_{\alpha\beta}\right) \\ &= \frac{1}{n^2} \left[\sum_{\alpha} \sum_{\beta} \text{Var}(y_{\alpha\beta}) + \sum_{\alpha, \alpha'} \sum_{\beta, \beta'} \text{Cov}(y_{\alpha\beta}, y_{\alpha'\beta'}) \right] \\ &= \frac{1}{n^2} \left[\sum_{\alpha} \sum_{\beta} \text{Var}(y_{\alpha\beta}) + \sum_{\alpha \neq \alpha'} \sum_{\beta} \sum_{\beta'} \text{Cov}(y_{\alpha\beta}, y_{\alpha'\beta'}) + \sum_{\alpha} \sum_{\beta \neq \beta'} \text{Cov}(y_{\alpha\beta}, y_{\alpha\beta'}) \right] = \\ &= \frac{1}{n^2} \left[\sum_{\alpha} \sum_{\beta} S^2 + \sum_{\alpha \neq \alpha'} \sum_{\beta} \sum_{\beta'} 0 + \sum_{\alpha} \sum_{\beta \neq \beta'} \rho S^2 \right] \\ &= \frac{1}{n^2} \left[nS^2 + a \cdot B(B-1)\rho S^2 \right] = \frac{S^2}{n} \left[1 + (B-1)\rho \right] \end{aligned}$$

Design effect: Intra-cluster homogeneity

- Hence

$$\begin{aligned} Deff &= \frac{Var(\bar{y})}{Var_{srs}(\bar{y})} \\ &= \frac{\frac{S^2}{n} [1 + (B-1)\rho]}{\frac{S^2}{n}} \\ &= [1 + (B-1)\rho] \end{aligned}$$

Features of ρ

- $-\frac{1}{(B-1)} < \rho < 1$ (although $\rho > 0$ generally)
- If $\rho = 0$, $Deff = 1.0$
 - The cluster sample is the equivalent of SRS of size $n = a \times B$
- If $\rho = 1$, $Deff = B$ and $Var(\bar{y}) = B \times Var_{srs}(\bar{y})$
 - The cluster sample is equivalent to an SRS of a elements

Source and nature of ρ

- For human populations in naturally occurring clusters, factors include
 - Environment (exposure to infectious disease)
 - Self-selection (poor households in same block)
 - Interaction (shared attitudes among neighbors)
- The size depends on
 - Characteristic Y (disease status, age)
 - Nature of clusters (naturally formed)
 - Size of cluster (blocks, census tracts, counties)

Estimation of ρ

- Proper estimation, especially for multistage stratified samples, is cumbersome
- It is useful for design to estimate ρ in a straightforward fashion
- Synthetic estimate:

$$roh = \frac{deff - 1}{B - 1}$$

Sample size considerations

- Implications of $\rho > 0$ for sample size:
 - Compute an SRS sample size (n_{srs})
 - For B and ρ (“guesstimate?”), compute $n = \left[1 + (B-1)\rho\right] n_{srs}$
- Design effect and confidence intervals:
 - With $(a-1)$ d.f.: $\bar{y} \pm \left(t_{(1-\alpha/2; a-1)}\right) se(\bar{y})$

or

$$\bar{y} \pm \left(t_{(1-\alpha/2; a-1)}\right) \sqrt{\left[1 + (B-1)\rho\right] \frac{s^2}{n}}$$

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- B. Sample size and *roh*
- C. Homogeneity and *roh*
- D. Portability of *roh*
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A. Subsampling clusters

- Select $b < B$ elements from a clusters *epsem* :

$$f = \frac{a}{A} \times \frac{b}{B} = \frac{a \times b}{A \times B} = \frac{n}{N}$$

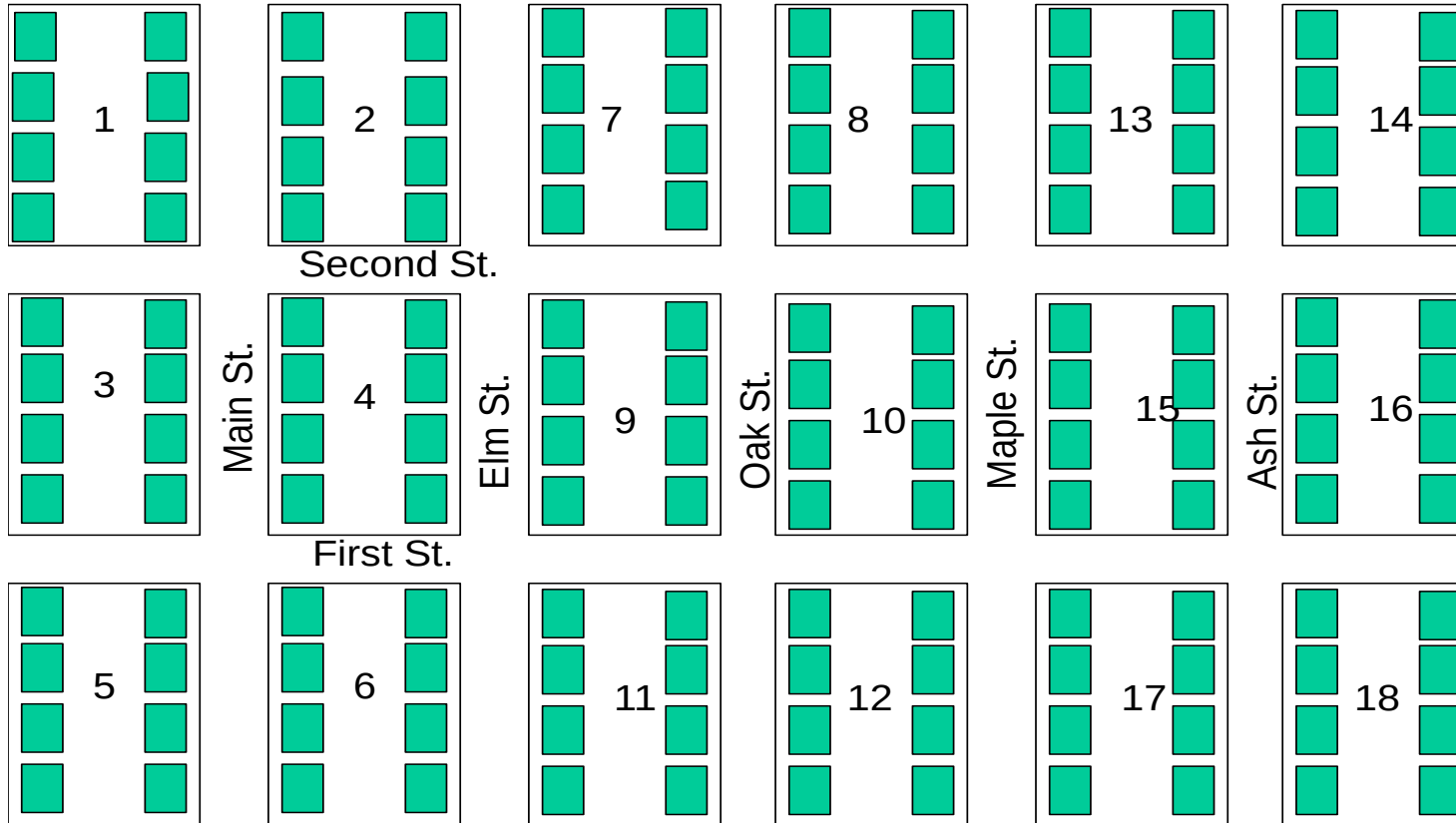
- SRS of clusters and SRS of elements within
- Define the average within cluster variance as $S_b^2 = \frac{1}{A} \sum_{\alpha=1}^A S_{\alpha}^2$
- Sampling variance changes:

$$Var(\bar{y}) = \left(1 - \frac{a}{A}\right) \frac{S_a^2}{a} + \left(1 - \frac{b}{B}\right) \frac{S_b^2}{ab}$$

Clustered population

$$\bar{Y}_\alpha = \sum_{\beta=1}^B Y_{\alpha\beta} / B$$

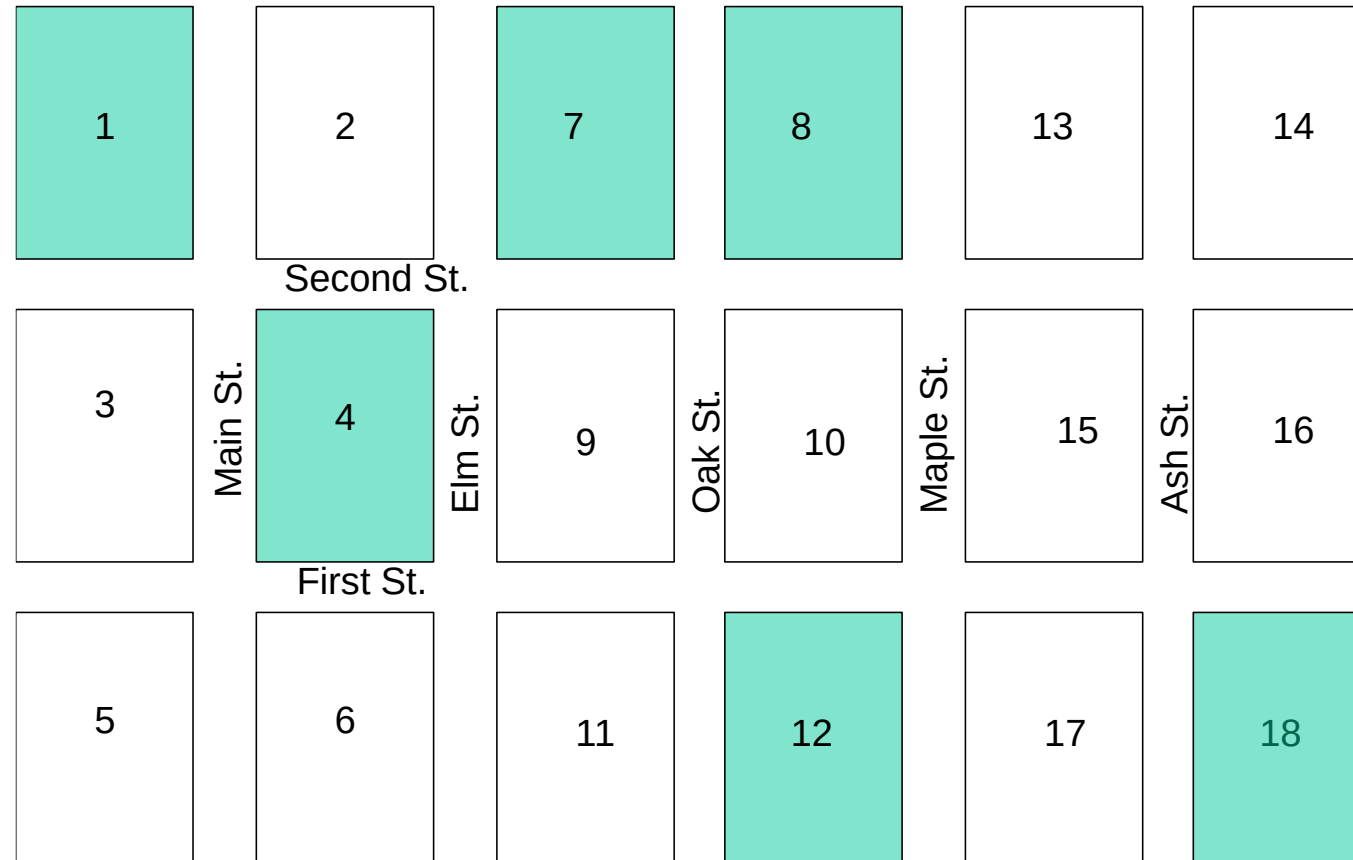
$$\bar{Y} = \sum_{i=1}^N Y_i / N$$



$$S_\alpha^2 = \sum_{\beta=1}^B (Y_{\alpha\beta} - \bar{Y}_\alpha)^2 / (B-1)$$

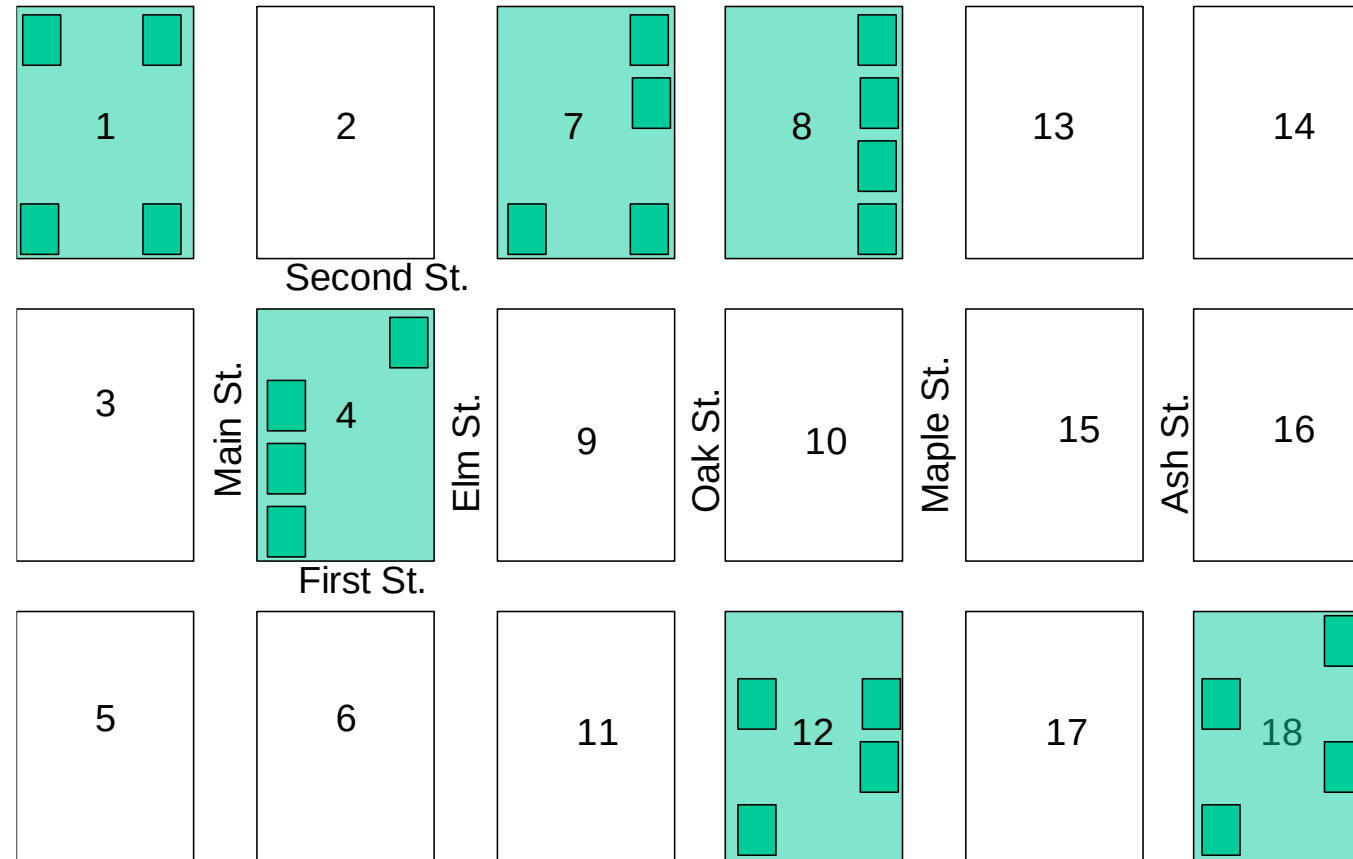
$$S^2 = \sum_{i=1}^N (Y_i - \bar{Y})^2 / (N-1)$$

Two-stage cluster sample: 1st stage



Two-stage cluster sample: 1st and 2nd stage

$$\bar{y} = \sum_{\alpha=1}^a \sum_{\beta=1}^b y_{\alpha\beta} / ab$$



$$Var(\bar{y}) = \left(1 - \frac{a}{A}\right) \frac{S_a^2}{a} + \left(1 - \frac{b}{B}\right) \frac{S_b^2}{ab}$$

Estimation of sampling variance

- Estimation of $Var(\bar{y})$ is complicated by the estimation of S_a^2

$$E(s_b^2) = S_b^2$$

$$E(s_a^2) = S_a^2 + \left(1 - \frac{b}{B}\right) \frac{S_b^2}{b}$$

$$\text{var}(\bar{y}) = \left(1 - \frac{a}{A}\right) \frac{s_a^2}{a} + \frac{a}{A} \left(1 - \frac{b}{B}\right) \frac{s_b^2}{ab}$$

Approximation

- If a/A is negligible, then $\text{var}(\bar{y})$ reduces to

$$\text{var}(\bar{y}) = \frac{1}{a} s_a^2$$

- This approximation avoids the need to compute s_b^2
- There is an alternative conceptual explanation for using a reduced or simplified variance estimate ...
- Ultimate clusters
- Leads to yet a second approximation: $\text{var}(\bar{y}) = \frac{(1-f)}{a} s_a^2$

Practical sampling variance estimation

- If clusters are selected with replacement and elements without, use the unbiased estimate:

$$\text{var}(\bar{y}) = \frac{(1-f)}{a} s_a^2$$

- If clusters are selected without replacement and element without, use a slightly biased estimate:

$$\text{var}(\bar{y}) \approx \frac{(1-f)}{a} s_a^2$$

instead of the more complicated

$$\text{var}(\bar{y}) = \left(1 - \frac{a}{A}\right) \frac{s_a^2}{a} + \left(\frac{a}{A}\right) \left(1 - \frac{b}{B}\right) \frac{s_b^2}{ab}$$

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B. Sample size and *roh*

- **Effective sample size**: size of SRS that would give the same precision as the cluster sample
- $n_{eff} = n/deff$
 - $deff = 2.0$ and $n = 1500$ then $n_{eff} = 1500/2 = 750$
- Varying subsample size for fixed n changes the number of clusters, $deff$, and the effective sample size
- For two-stage cluster sample: $deff = 1 + (b - 1)roh$

Effective sample size

- Cluster sample with $n = 2000$, and $roh = 0.03$:
- For $b = 10$, $deff = 1 + (10 - 1)0.03 = 1.27$
 - $n_{eff} = 2000/1.27 = 1574.80$
- For $b = 20$, $deff = 1 + (20 - 1)0.03 = 1.57$
 - $n_{eff} = 2000/1.57 = 1273.89$
- For $b = 50$, $deff = 1 + (50 - 1)0.03 = 2.47$
 - $n_{eff} = 2000/2.47 = 809.72$

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C. Homogeneity and *roh*

- Sample 10 school classrooms from 1,000
 - Each classroom has exactly 24 children
- Alternative 1: sample characteristic is dichotomy

$$\frac{9}{24}, \frac{11}{24}, \frac{13}{24}, \frac{15}{24}, \frac{16}{24}, \frac{17}{24}, \frac{18}{24}, \frac{20}{24}, \frac{20}{24}, \frac{21}{24}$$

- Here the intra-class correlation $roh = 0.088$
- Moderate amount of homogeneity within clusters
- $deff = 1 + (24 - 1)0.088 = 3.024$
- Actual sample size $n = 240$
- Effective sample size $n_{eff} = 240/3.024 = 79.37$

Perfect homogeneity

- Homogeneity within, heterogeneity among:

$$\frac{0}{24}, \frac{0}{24}, \frac{0}{24}, \frac{16}{24}, \frac{24}{24}, \frac{24}{24}, \frac{24}{24}, \frac{24}{24}, \frac{24}{24}, \frac{24}{24}$$

$$deff = 23.90$$

$$roh = \frac{23.90 - 1}{24 - 1} = 0.996$$

$$n_{eff} = 240 / 23.9 = 10.04$$

Perfect heterogeneity

- Heterogeneity within, homogeneity among:

$$\frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}, \frac{16}{24}$$

$$deff = 0$$

$$roh = \frac{0-1}{24-1} = -0.04348$$

$$n_{eff} = 240 / 0 = \text{undefined (total population)}$$

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- C. Homogeneity and *roh*
- D. Portability of *roh*
- E. Subsample size

D. Portability of *roh*

- Estimation (1)

$$\text{var}_{(1)}(p) = \frac{(1-f)}{a} s_a^2$$

$$\text{var}_{(1),SRS}(\bar{y}) = \frac{p(1-p)}{n_{(1)}}$$

$$\text{deff}_{(1)} = \frac{\text{var}_{(1)}(p)}{\text{var}_{(1),SRS}(p)}$$

$$\text{roh} = \frac{\text{deff}_{(1)} - 1}{b_{(1)} - 1}$$

- Design (2)

$$\text{var}_{(2)}(p) = \text{deff}_{(2)} \times \text{var}_{(2),SRS}(p)$$

$$\text{var}_{(2),SRS}(\bar{y}) = \frac{p(1-p)}{n_{(2)}}$$

$$\text{deff}_{(2)} = 1 + (b_{(2)} - 1)\text{roh}$$

roh

Illustration

- Crime victimization survey in a large public housing project
- Sample apartments
- Responsible adult interviewed about victimizations occurring to any member of the household
- $A = 400$ floors with exactly $B = 15$ apartments
- $a = 10$ floors selected with replacement
- $b = 5$ apartments selected without replacement

Sample results

$$f = \frac{a}{A} \times \frac{b}{B} = \frac{10}{400} \times \frac{5}{15} = \frac{50}{6000} = \frac{1}{120}$$

Sample floor	HH's experienced any crime (y_α)
1	4
2	4
3	5
4	5
5	3
6	1
7	0
8	1
9	2
10	1

Estimation

$$p = \bar{y} = \sum_{\alpha=1}^a \sum_{\beta=1}^b y_{\alpha\beta} / n = 26 / 50 = 0.52$$

$$\text{var}(\bar{y}) = \frac{1-f}{a} s_a^2$$

$$s_a^2 = \frac{1}{b^2(a-1)} \left[\sum_{\alpha=1}^a y_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a y_{\alpha} \right)^2}{a} \right] = \frac{1}{(5)^2(9)} \left[98 - \frac{(26)^2}{10} \right] = 0.13511$$

$$\text{var}(p) = \frac{1 - \frac{1}{120}}{10} (0.13511) = 0.013399$$

$$se(p) = 0.11575$$

Confidence interval, *deff* and *roh*

$$t_{(0.975, a-1)} = t_{(0.975, 9)} = 2.262$$

$$p \pm t_{(0.975, 9)} se(p) = (0.25815, 0.78185)$$

$$\begin{aligned} \text{var}_{srs}(p) &= (1-f) \frac{p(1-p)}{n-1} = \\ &= \left(1 - \frac{1}{120}\right) \frac{(0.52)(0.48)}{49} = 0.005051 \end{aligned}$$

$$deff = \frac{\text{var}(p)}{\text{var}_{srs}(p)} = \frac{0.013399}{0.0050514} = 2.6524$$

$$roh = \frac{deff - 1}{b - 1} = \frac{2.6524 - 1}{5 - 1} = 0.41311$$

Unit 5: Clusters

Two-Stage Cluster Sampling

- A. Subsampling clusters
- B. Sample size and *roh*
- C. Homogeneity and *roh*
- D. Portability of *roh*
- E. Subsample size

Unit 5: Clusters

Two-Stage Cluster Sampling

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E. Subsample size

- Precision of estimates depends on a and b
- Optimum choice for a and b can be obtained by minimizing the sampling variance for fixed cost (or vice versa)
- Cost model for two stage sampling:
 - $C - C_0 = a \times c_a + a \times (b \times c_b)$
 - C_a is the cost per cluster and C_b is the cost per observation within a cluster
- C_b is dominated by interviewing costs
- C_a is dominated by travel and preparation costs

Optimization

- Optimum subsample size for fixed cost $C - C_0$ can be found using a Lagrange multiplier or the Cauchy-Schwartz inequality
- Find a (or b) to minimize the sampling variance

$$Var(\bar{y}) = \frac{S_a^2}{a} + \frac{S_b^2}{ab}$$

- The optimum b is

$$b_{opt} = \sqrt{\frac{c_a}{c_b} \cdot \frac{1 - roh}{roh}}$$

Illustration

- Large national personal visit interview survey
- Interviews conducted within respondent homes by locally hired interviewers
- *roh* values near 0.05
- $n = 1,000$ is fixed
- 1 hour interview requires 2 hours interviewer time
 - Interviewer travels to the cluster, locating households, making callbacks, etc.
 - At \$12.50 per interviewer hour, $c_b = 2 \times \$12.50 = \25

Cost per cluster

- Household clusters (segments) an average 1/2 hour drive from interviewer homes
- Require an average of 3 visits each
- $3 \times \$12.50 = \37.50 required to pay interviewers for travel time
- Travel averages 30 m.p.h.
- At \$0.31 per mile, 6 trips of one half hour (15 miles) requires travel reimbursement of $(6 \times 15)(0.31) = \$27.90$
- $c_a = 37.50 + 27.90 = \$65.40$

Optimum b

- The optimum subsample size will be

$$b_{opt} = \sqrt{\frac{65.40}{25} \cdot \frac{1-0.05}{0.05}} = 7.05$$

Unit 6: Unequal Sized Clusters

Unequal Sized Cluster Sampling

- A. Unequal sized clusters
- B. Ratio means
- C. Illustration

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A. Unequal sized clusters

- Unequal-sized clusters (with subsampling) pose problems for sample selection and estimation
- Fixed two-stage sampling rates $f_a=1/2$ and $f_b=1/10$ (i.e., $f=1/20$) for the following population of $A = 10$ unequal sized clusters with $N=1850$:

$$f = f_a \times f_b = \frac{1}{2} \times \frac{1}{10} = \frac{1}{20} \text{ and } f \times N = \frac{1}{20} \times 1850 = 92.5$$

Two unequal cluster samples, $f=1/20$

Cluster α	Size B_α	Sample 1			Sample 2		
		α	B_α	$x_\alpha = f_b B_\alpha$	α	B_α	$x_\alpha = f_b B_\alpha$
1	400	1	400	40	2	310	31
2	310						
3	40	3	40	4	4	150	15
4	150						
5	250	5	250	25	6	220	22
6	220						
7	50	7	50	5	8	130	13
8	130						
9	90	9	90	9	10	210	21
10	210						
Total	1850	5	830	83	5	1020	102

Variation in sample size

- Sample size is a random variable
 - y/n is no longer appropriate
 - Ratio mean $r=y/x$ is appropriate
- Alternatives to variation in sample size:
 - Depart from *epsem*
 - Leads to weighting
 - Increase in variance due to weights
 - Create new clusters of equal size
 - Stratify by size
 - Probability proportional to size selection (PPS) [We'll come back to this later]

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B. Ratio means

- Two principle problems with ratio means:
 - Biased for the overall population mean
 - The sampling variance of the ratio mean is not known exactly (except for some special sample designs)
 - Estimate sampling variance using Taylor Series Linearization:

$$r = \frac{y}{x} = \frac{\sum_{\alpha=1}^a y_{\alpha}}{\sum_{\alpha=1}^a x_{\alpha}} = \frac{\sum_{\alpha=1}^a \sum_{\beta=1}^{x_{\alpha}} y_{\alpha\beta}}{\sum_{\alpha=1}^a x_{\alpha}}$$

$$\text{var}(r) \approx \frac{1}{x^2} \left[\text{var}(y) + r^2 \text{var}(x) - 2r \text{cov}(y, x) \right]$$

Variance for sample total

- Recall from equal size one-stage cluster sampling,

$$\text{var}(\bar{y}) = \frac{1-f}{aB^2(a-1)} \left(\sum_{\alpha=1}^a y_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a y_{\alpha} \right)^2}{a} \right)$$

- For sample total,

$$\begin{aligned} \text{var}(y) &= \text{var}(aB \times \bar{y}) = (aB)^2 \text{var}(\bar{y}) \\ &= a \frac{1-f}{a-1} \left(\sum_{\alpha=1}^a y_{\alpha}^2 - \frac{y^2}{a} \right) \end{aligned}$$

Estimating sample total variances

- For an SRS of a clusters

$$\text{var}(y) = a \frac{(1-f)}{a-1} \left(\sum_{\alpha=1}^a y_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a y_{\alpha} \right)^2}{a} \right)$$

$$\text{var}(x) = a \frac{(1-f)}{a-1} \left(\sum_{\alpha=1}^a x_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a x_{\alpha} \right)^2}{a} \right)$$

$$\text{cov}(y, x) = a \frac{(1-f)}{a-1} \left(\sum_{\alpha=1}^a y_{\alpha} x_{\alpha} - \frac{\left(\sum_{\alpha=1}^a y_{\alpha} \right) \left(\sum_{\alpha=1}^a x_{\alpha} \right)}{a} \right)$$

Estimating $\text{var}(r)$

- Substituting into $\text{var}(r)$, and simplifying,

$$\text{var}(r) \approx \frac{1}{x^2} \left[\text{var}(y) + r^2 \text{var}(x) - 2r \text{cov}(y, x) \right] =$$

$$= \left(\frac{1}{x^2} \right) a \frac{(1-f)}{(a-1)} \left[\left(\sum_{\alpha=1}^a y_{\alpha}^2 - \frac{y^2}{a} \right) + r^2 \left(\sum_{\alpha=1}^a x_{\alpha}^2 - \frac{x^2}{a} \right) - 2r \left(\sum_{\alpha=1}^a y_{\alpha} x_{\alpha} - \frac{yx}{a} \right) \right]$$

$$= \left(\frac{1}{x^2} \right) a \frac{(1-f)}{(a-1)} \left[\sum_{\alpha=1}^a y_{\alpha}^2 + r^2 \sum_{\alpha=1}^a x_{\alpha}^2 - 2r \sum_{\alpha=1}^a y_{\alpha} x_{\alpha} \right]$$

Adequacy of approximation

- Since $\text{var}(r)$ is an approximation, it would be useful to an indication of when it might fail
- Examination of the Taylor series shows the adequacy of the approximation depends on the coefficient of variation of the sample size:

$$cv(x) = \frac{se(x)}{x} = \frac{\sqrt{\text{var}(x)}}{x} = \frac{\sqrt{a \frac{(1-f)}{a-1} \left(\sum_{\alpha=1}^a x_{\alpha}^2 - \frac{\left(\sum_{\alpha=1}^a x_{\alpha} \right)^2}{a} \right)}}{\sum_{\alpha=1}^a x_{\alpha}}$$

Rule of thumb for adequacy and bias

- As long as $cv(x) < 0.15$, the approximation is reasonably accurate
 - Values of $cv(x)$ as large as 0.20 may also be acceptable
- The bias of r also depends on the coefficient of variation of the denominator:

$$\left| \frac{Bias(r)}{se(r)} \right| < cv(x)$$

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C. Illustration: Ratio mean

- In a hospital authority of 5 hospitals, estimate the proportion of outpatient visits due to trauma
 - Element: outpatient visits
 - Estimate: proportion of visits due to trauma
- Select outpatient records in 2 stages:
 - First hospital-days ($A = 5 \times 365 = 1825$)
 - Clusters: days across hospitals
 - Sample of $a = 10$ hospital days
 - Second, select all records on a selected day

$$f = \frac{10}{1825} \cdot 1 = \frac{10}{1825}$$

Hospital-days

Sample hospital-day α	Total visits x_α	Trauma visits y_α
1	58	40
2	47	16
3	37	8
4	69	27
5	40	10
6	27	18
7	34	17
8	30	12
9	26	16
10	32	16

$$\sum_{\alpha=1}^a x_\alpha = 400 \quad \sum_{\alpha=1}^a y_\alpha = 180 \quad \sum_{\alpha=1}^a x_\alpha^2 = 17778 \quad \sum_{\alpha=1}^a y_\alpha^2 = 4018 \quad \sum_{\alpha=1}^a x_\alpha y_\alpha = 7983$$

Inference

- For $r = 0.45$, estimate $\text{var}(r)$, ignoring the fpc:

$$\begin{aligned}\text{var}(r) &\approx \frac{1}{x^2} \frac{a}{a-1} \left[\sum_{\alpha=1}^a y_{\alpha}^2 + r^2 \sum_{\alpha=1}^a x_{\alpha}^2 - 2r \sum_{\alpha=1}^a y_{\alpha} x_{\alpha} \right] = \\ &= \frac{1}{400^2} \frac{10}{10-1} \left[4018 + 0.45^2 \times 17788 - 2 \times 0.45 \times 7983 \right] = 0.003023\end{aligned}$$

- For $t_{(0.975, 9)} = 2.262$, 95% C.I. (0.3256, 0.5744)
- Adequacy of the approximation:

$$cv(x) = \frac{\sqrt{\frac{a}{a-1} \left(\sum_{\alpha=1}^a x_{\alpha}^2 - \frac{x^2}{a} \right)}}{\sum_{\alpha=1}^a x_{\alpha}} = \frac{\sqrt{\frac{10}{10-1} \left(17788 - \frac{400^2}{a} \right)}}{400} = 0.1114$$

Design of new sample

- *deff* and *roh*:

$$\text{var}_{\text{SRS}}(r) = \frac{r(1-r)}{n-1} = \frac{(0.45)(0.55)}{399} = 0.0006203$$

$$\text{deff} = \frac{\text{var}(r)}{\text{var}_{\text{SRS}}(r)} = \frac{0.003023}{0.0006203} = 4.874$$

$$\text{roh} = \frac{\text{deff} - 1}{\bar{b} - 1} = \frac{4.874 - 1}{40 - 1} = 0.09934$$

$$\text{where } \bar{b} = \frac{400}{10} = 40 \left(= \frac{x}{a} \right)$$

- What would happen if $a = 10$ and $\bar{b} = 20$?
 - Compute new *deff*, and multiply by SRS sampling variance

Unit 6: Unequal Sized Clusters

Stratified Unequal Sized Cluster Sampling

- A. Principles of selection
- B. Variance estimation

Unit 6: Unequal Sized Clusters

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A. Principles of selection

- Cluster sample stratification similar to elements
 - Use cluster characteristics
 - Homogeneous, mutually exclusive, exhaustive
 - Control the distribution of the sample
 - Decrease sampling variance
 - Selection of stratifying variables, boundaries, etc., discussed for element sampling applies to clusters
- Allocation of sample clusters across strata:
 - Proportionate allocation
 - Paired selection
 - Other allocations

Notation

- Add subscript to denote cluster stratification
- Population of N elements, $B_{h\alpha}$ per cluster
- A_h clusters per stratum
- Population mean: $\bar{Y} = \sum_{h=1}^H \sum_{\alpha=1}^{A_h} \sum_{\beta=1}^{B_{h\alpha}} Y_{h\alpha\beta} / N$

Sample estimation

- At the sample level, $y_{h\alpha\beta}$ denotes survey variable values
- Selection of a_h from A_h clusters (with or without replacement)
- Selection of $b_{h\alpha}$ elements from $(h\alpha)^{th}$ cluster
- Simple replicated subsampling, or ultimate cluster sampling, within each stratum

$$r = \frac{y}{x} = \frac{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{\beta=1}^{b_{h\alpha}} y_{h\alpha\beta}}{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{\beta=1}^{b_{h\alpha}} x_{h\alpha\beta}} = \frac{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} y_{h\alpha}}{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} x_{h\alpha}} = \frac{\sum_{h=1}^H y_h}{\sum_{h=1}^H x_h}$$

Weighted estimation

- The quantities $y_{h\alpha}$ and $x_{h\alpha}$ can be weighted, if not *epsem*
 - r implicitly uses $y_{h\alpha\beta}^* = w_{h\alpha\beta} y_{h\alpha\beta}$
 - $w_{h\alpha\beta}$ is the weight for the $(h\alpha\beta)^{th}$ element
 - $y_{h\alpha} = \sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta} y_{h\alpha\beta} = \sum_{\beta=1}^{b_{h\alpha}} y_{h\alpha\beta}^*$ is the weighted estimate for the primary selection in stratum h
- “Best” estimate of the total $Y_{h\alpha}$
- Similarly, $x_{h\alpha\beta}^* = w_{h\alpha\beta} x_{h\alpha\beta}$ and $x_{h\alpha} = \sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta} x_{h\alpha\beta} = \sum_{\beta=1}^{b_{h\alpha}} x_{h\alpha\beta}^*$

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B. Variance estimation

- Sampling variance estimation depends on the methods used to select the clusters within each stratum
 - Based on ultimate cluster sampling theory
 - Under disproportionate allocation, $(1 - f_h)$ appears in each stratum
- Three different methods
 - Multiple differences
 - Paired selection
 - Successive differences

Multiple differences

$$\begin{aligned}\text{var}(r) &\approx \frac{1}{x^2} \left[\text{var}(y) + r^2 \text{var}(x) - 2r \text{cov}(y, x) \right] \\ &= \frac{1}{x^2} \left[\sum_{h=1}^H \text{var}(y_h) + r^2 \sum_{h=1}^H \text{var}(x_h) - 2r \sum_{h=1}^H \text{cov}(y_h, x_h) \right] \\ &= \frac{1}{x^2} \left[\sum_{h=1}^H \frac{a_h (1 - f_h)}{a_h - 1} \left(\sum_{\alpha=1}^{a_h} y_{h\alpha}^2 - \frac{y_h^2}{a_h} \right) \right. \\ &\quad \left. + r^2 \sum_{h=1}^H \frac{a_h (1 - f_h)}{a_h - 1} \left(\sum_{\alpha=1}^{a_h} x_{h\alpha}^2 - \frac{x_h^2}{a_h} \right) \right. \\ &\quad \left. - 2r \sum_{h=1}^H \frac{a_h (1 - f_h)}{a_h - 1} \left(\sum_{\alpha=1}^{a_h} y_{h\alpha} x_{h\alpha} - \frac{y_h x_h}{a_h} \right) \right]\end{aligned}$$

Paired selection

- $a_h = 2$ for all h

$$\text{var}(r) \approx \frac{1}{x^2} \left[\sum_{h=1}^H (1-f_h)(y_{h1} - y_{h2})^2 + r^2 \sum_{h=1}^H (1-f_h)(x_{h1} - x_{h2})^2 - 2r \sum_{h=1}^H (1-f_h)(y_{h1} - y_{h2})(x_{h1} - x_{h2}) \right]$$

- Selection may be
 - Paired selection design
 - Systematic selection collapsed to paired differences
 - One selection per stratum collapsed to pairs

Successive differences

- Systematic selection from ordered list
- Successive differences ($a_h > 2$ for all h):

$$\text{var}(r) \approx \frac{1}{x^2} \left[\begin{aligned} &\sum_{h=1}^H \frac{a_h (1-f_h)}{2(a_h-1)} \sum_{g=1}^{a_h-1} (y_{hg} - y_{h,g+1})^2 \\ &+ r^2 \sum_{h=1}^H \frac{a_h (1-f_h)}{2(a_h-1)} \sum_{g=1}^{a_h-1} (x_{hg} - x_{h,g+1})^2 \\ &- 2r \sum_{h=1}^H \frac{a_h (1-f_h)}{2(a_h-1)} \sum_{g=1}^{a_h-1} (y_{hg} - y_{h,g+1})(x_{hg} - x_{h,g+1}) \end{aligned} \right]$$

Alternative formulation

- These formulas can also be expressed in terms of z_{ha}
- Reflect an underlying bivariate regression of y on x through the origin
- $z_{h\alpha} = y_{h\alpha} - rx_{h\alpha}$

Hospital example revisited

Sample hospital-day α	Total visits x_α	Trauma visits y_α	$z_\alpha = y_\alpha - rx_\alpha$
1	58	40	13.9
2	47	16	-5.2
3	37	8	-8.7
4	69	27	-4.1
5	40	10	-8.0
6	27	18	5.9
7	34	17	1.7
8	30	12	-1.5
9	26	16	4.3
10	32	16	1.6

Three formulas revisited

$$\text{var}(r) \approx \frac{1}{x^2} \left[\sum_{h=1}^H \frac{a_h (1 - f_h)}{a_h - 1} \left(\sum_{\alpha=1}^{a_h} z_{h\alpha}^2 - \frac{z_h^2}{a_h} \right) \right]$$

$$\text{var}(r) \approx \frac{1}{x^2} \left[\sum_{h=1}^H (1 - f_h) (z_{h1} - z_{h2})^2 \right]$$

$$\text{var}(r) \approx \frac{1}{x^2} \left[\sum_{h=1}^H \frac{a_h (1 - f_h)}{2(a_h - 1)} \sum_{g=1}^{a_h-1} (z_{hg} - z_{h,g+1})^2 \right]$$

Unit 6: Unequal Sized Clusters

Complex Designs

- A. Illustration
- B. Probabilities of selection
- C. Weighted mean
- D. Increase in variance

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A. Illustration

- Select adults in public housing apartments
- Study problem of crime victimization among public housing residents
- Select a sample of $a = 10$ apartments from master list provided by the public housing authority
 - Among $A_1 = 300$ apartments east of the expressway, select $a_1 = 5$ apartments and adults
 - Among $A_2 = 100$ apartments west of the expressway, select $a_2 = 5$ apartments and adults
- Within apartments, list adults, select 1 at random
- Obtain a report of the number of times selected adult was victimized in the last 12 months

Sample data

Stratum h	Apt. α	No. adults $B_{h\alpha}$	Resp. $y_{h\alpha\beta}$	Weight $w_{h\alpha\beta}$	Wgtd. Resp. $w_{h\alpha\beta} y_{h\alpha\beta}$
East	1	2	2	120	240
	2	1	0	60	0
	3	3	1	180	180
	4	2	4	120	480
	5	1	2	60	120
West	1	1	0	20	0
	2	2	1	40	40
	3	2	1	40	40
	4	3	5	60	300
	5	2	2	40	80

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B. Probabilities of selection

- Probabilities of selection vary by strata and apartments:

$$\Pr\{h\alpha\beta\} = \Pr\{h\alpha\} \times \Pr\{\beta \mid h\alpha\}$$

- h denotes the "side" of the expressway
- α denotes the apartment
- β denotes the adult within the apartment

- Alternatively,

$$\Pr\{\text{selecting adult}\} = \Pr\{\text{selecting apartment}\} \times \Pr\{\text{selecting adult} \mid \text{apartment selected}\}$$

- Compensatory weight the inverse: $w_{h\alpha\beta} = 1/\Pr\{h\alpha\beta\}$

Sample data

Stratum h	Apt. α	No. adults $B_{h\alpha}$	Resp. $y_{h\alpha\beta}$	Weight $w_{h\alpha\beta}$	Wgtd. Resp. $w_{h\alpha\beta} y_{h\alpha\beta}$
East	1	2	2	120	240
	2	1	0	60	0
	3	3	1	180	180
	4	2	4	120	480
	5	1	2	60	120
West	1	1	0	20	0
	2	2	1	40	40
	3	2	1	40	40
	4	3	5	60	300
	5	2	2	40	80

Two apartments

- For an east-side apartment, $\Pr\{11\beta\} = \Pr\{11\} \times \Pr\{\beta | 11\} = \frac{5}{300} \times \frac{1}{B_{11}}$
 - B_{11} = number of adults in apartment $\{11\} = 2$
 - $\Pr\{11\beta\} = \frac{5}{300} \times \frac{1}{2} = \frac{1}{120}$
 - Weight $w_{11\beta} = (1/120)^{-1} = 120$
- Similarly, $\Pr\{21\beta\} = \Pr\{21\} \times \Pr\{\beta | 21\} = \frac{5}{100} \times \frac{1}{B_{21}}$
- Weight $w_{21\beta} = (1/20)^{-1} = 20$

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- D. Increase in variance

C. Weighted mean

- The weighted estimator of the mean is

$$\bar{y}_w = \frac{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta} y_{h\alpha\beta}}{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta}} = \frac{1480}{740} = 2.0$$

- In contrast, the unweighted estimate is

$$\bar{y} = \frac{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{\beta=1}^{b_{h\alpha}} y_{h\alpha\beta}}{n} = \frac{18}{10} = 1.8$$

Variance estimation

- Variance estimation takes into account the fact that the denominator varies
- Depends on which sample of apartments selected
- Weighted mean is a ratio of random variables:

$$\bar{y}_w = \frac{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta} y_{h\alpha\beta}}{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} \sum_{\beta=1}^{b_{h\alpha}} w_{h\alpha\beta}} = \frac{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} u_{h\alpha}}{\sum_{h=1}^H \sum_{\alpha=1}^{a_h} w_{h\alpha}} = \frac{u}{w}$$

Taylor series approximation

- The approximation is

$$\text{var}(\bar{y}_w) \approx \frac{1}{w^2} \left[\text{var}(u) + r^2 \text{var}(w) - 2r \text{cov}(u, w) \right]$$

- For stratified clusters, ignoring the fpc,

$$\text{var}(u) = \sum_{h=1}^H \left(\frac{a_h}{a_h - 1} \right) \left(\sum_{\alpha=1}^{a_h} u_{h\alpha}^2 - \frac{u_h^2}{a_h} \right)$$

$$\text{var}(w) = \sum_{h=1}^H \left(\frac{a_h}{a_h - 1} \right) \left(\sum_{\alpha=1}^{a_h} w_{h\alpha}^2 - \frac{w_h^2}{a_h} \right)$$

$$\text{cov}(w, u) = \sum_{h=1}^H \left(\frac{a_h}{a_h - 1} \right) \left(\sum_{\alpha=1}^{a_h} w_{h\alpha} u_{h\alpha} - \frac{w_h u_h}{a_h} \right)$$

Sample data

Stratum	Apt. α	Wght. $w_{h\alpha}$	$u_{h\alpha}$	$w_{h\alpha}^2$	$u_{h\alpha}^2$	$w_{h\alpha} u_{h\alpha}$
East	1	120	240	14400	57600	28800
	2	60	0	3600	0	0
	3	180	180	32400	32400	32400
	4	120	480	14400	230400	57600
	5	60	120	3600	14400	7200
Subtotal		540	1020	68400	334800	126000
West	1	20	0	400	0	0
	2	40	40	1600	1600	1600
	3	40	40	1600	1600	1600
	4	60	300	3600	90000	18000
	5	40	80	1600	6400	3200
Subtotal		200	460	8800	99600	24400
Total	10	740	1480	77200	434400	150400

Variance of sample totals

$$\begin{aligned}\text{var}(u) &= \sum_{h=1}^H \left(\frac{a_h}{a_h - 1} \right) \left(\sum_{\alpha=1}^{a_h} u_{h\alpha}^2 - \frac{u_h^2}{a_h} \right) \\ &= \left(\frac{5}{4} \right) \left(334800 - \frac{1020^2}{5} \right) + \left(\frac{5}{4} \right) \left(99600 - \frac{460^2}{5} \right) = 230,000\end{aligned}$$

$$\begin{aligned}\text{var}(w) &= \sum_{h=1}^H \left(\frac{a_h}{a_h - 1} \right) \left(\sum_{\alpha=1}^{a_h} w_{h\alpha}^2 - \frac{w_h^2}{a_h} \right) \\ &= \left(\frac{5}{4} \right) \left(68400 - \frac{540^2}{5} \right) + \left(\frac{5}{4} \right) \left(8800 - \frac{200^2}{5} \right) = 13,600\end{aligned}$$

$$\begin{aligned}\text{cov}(u, w) &= \sum_{h=1}^H \left(\frac{a_h}{a_h - 1} \right) \left(\sum_{\alpha=1}^{a_h} u_{h\alpha} w_{h\alpha} - \frac{u_h w_h}{a_h} \right) \\ &= \left(\frac{5}{4} \right) \left(126000 - \frac{540 \times 1020}{5} \right) + \left(\frac{5}{4} \right) \left(24400 - \frac{200 \times 460}{5} \right) = 27,300\end{aligned}$$

Ratio mean estimated variance

- Combining these estimates,

$$\begin{aligned}\text{var}(\bar{y}_w) &\approx \frac{1}{740^2} \left[230,000 + (2.0)^2 (13,600) - 2(2.0)(27,300) \right] \\ &= 0.3199\end{aligned}$$

$$se(\bar{y}_w) = 0.5656$$

- For confidence intervals, $df = a - H$

Unit 6: Unequal Sized Clusters

Complex Designs

- A. Illustration
- B. Probabilities of selection
- C. Weighted mean
- D. Increase in variance

Unit 6: Unequal Sized Clusters

Complex Designs

- A. Illustration
- B. Probabilities of selection
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- D. Increase in variance

D. Increase in variance

- Weighting can increase sampling variance of estimates
- Assuming “arbitrary” weighting
- Potential increase in variance due to weighting:

$$1 + L = \frac{n \left(\sum_{i=1}^n w_i^2 \right)}{\left(\sum_{i=1}^n w_i \right)^2} = \frac{10(77200)}{(740)^2} = 1.410$$

- Increase depends on variability of weights.

Unit 6: Unequal Sized Clusters

PPS

- A. Controlling sample size
- B. PPS
- C. Systematic PPS

Unit 6: Unequal Sized Clusters

PPS

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A. Controlling sample size

- Potentially large variation in sample size
 - A ratio mean for design-consistent estimation
- More control over sample size. Why?
 - Bias of r
 - Variance of r
- Equalize workloads controlling survey costs

Example: Two-stage selection

- A sample of $n = 36$ employees from $N = 2,160$
 - $f = 36/2160 = 1/60$

Unit	No.	Unit	No.
1	443	6	291
2	162	7	64
3	127	8	70
4	554	9	232
5	115	10	102

Sample size control

- Redefine clusters
- Stratification by size, varying sampling rates by size of cluster stratum
- Give up *epsem* for equal size subsamples
- Probabilities proportionate to size (PPS)

Redefine clusters

- Create 10 clusters of approximately $N/10 = 216$ employees each
- Variation above and below desired cluster size
- Once clusters created, select the sample with uniform sampling fractions

$$f = f_a f_b = \left(\frac{a}{A}\right)\left(\frac{b}{B}\right)$$

$$fN = \left(\frac{1}{60}\right)(2160) = 36$$

Stratify by size

Stratum I		Stratum II	
Unit	No.	Unit	No.
2	162	1	443
3	127	4	554
5	115	6	291
7	64	9	<u>232</u>
8	70		1520
10	<u>102</u>		
	640		

Varying sampling rates

- Obtain *epsem* by choice of first and second stage rates:
- Use in stratum I: $f_{Ia} = \frac{1}{6}$ $f_{Ib} = \frac{1}{10}$
- Use in stratum II: $f_{IIa} = \frac{1}{4}$ $f_{IIb} = \frac{1}{15}$
- Sample size in stratum I ranges from 6 to 17, stratum II from 15 to 37
- $cv(x)$ has been reduced within strata.

Equal subsample size

- Select clusters with equal rates but take equal size subsamples
- Non-*epsem* design

$$f = f_a f_{b|a} = \left(\frac{a}{A}\right)\left(\frac{b}{B_\alpha}\right) = \left(\frac{2}{10}\right)\left(\frac{18}{B_\alpha}\right) = \frac{16}{10 \times B_\alpha} = f_\alpha$$

- Knowing B_α for each selected cluster, create weights $w_{\alpha\beta} = f_\alpha^{-1}$
- Constant overall workloads at the price of weighting and increases in sampling variances relative to *epsem* designs

Unit 6: Unequal Sized Clusters

PPS

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Unit 6: Unequal Sized Clusters

PPS

- A. Controlling sample size
- B. PPS**
- C. Systematic PPS

PPS, or how to achieve *epsem* and equal subsample sizes at the same time

- Goal: Select a two stage sample such that there are equal subsample sizes in each cluster with *epsem*
- We want an overall *epsem* rate $f = n/N$ such that the sampling rate in the second stage is b/B_α

$$f = f_\alpha f_{\beta|\alpha} = \left(f_\alpha \right) \left(\frac{b}{B_\alpha} \right) = \left(\frac{n}{N} \right)$$

- Solving $f_\alpha = \frac{n}{N} \times \frac{B_\alpha}{b} = \frac{ab}{\sum_{\alpha=1}^A B_\alpha} \times \frac{B_\alpha}{b} = \frac{aB_\alpha}{\sum_{\alpha=1}^A B_\alpha}$

PPS implications

- Select clusters with probabilities proportionate to their size, B_α
 - Overall the selection becomes $f = \frac{aB_\alpha}{\sum_{\alpha=1}^A B_\alpha} \times \frac{b}{B_\alpha}$
- Selection probability varies at both first & second stages
 - Overall, *epsem*
 - Large clusters selected with **high probabilities**, *subsampling* at **low rate**
 - Small clusters selected with **low probabilities**, *subsampling* at **high rate**

Selecting clusters with PPS

- Two approaches
 - Simple replicated subsampling – at random and with replacement
 - Systematic sample and without replacement
- For both approaches, need to cumulate (sequentially add up) the size of the clusters

Selecting clusters with PPS

Unit	B_{α}	Cum. B_{α}	Unit	B_{α}	Cum. B_{α}
1	443	443	6	291	1692
2	162	605	7	64	1756
3	127	732	8	70	1826
4	554	1286	9	232	2058
5	115	1401	10	102	2160

Using cumulative counts

- Simple replicated subsampling for $a=2$:
 - Select RN from 1 to 2160 with replacement
 - Find the cluster (Unit) with cumulative sum first exceeding RN
 - Select
 - Repeat
 - Subsampling rate b / B_α requires exactly b
- If cluster selected k times, take $k \times b$ elements
- Subsampling: SRS, systematic, stratified element, even cluster sampling

Calculating subsampling rates

- For example, $RN_1=702$ and $RN_2 = 1744$ select Units 3 and 7
- For Unit 3, $\frac{aB_\alpha}{N} = \frac{2 \times 127}{2160} = \frac{1}{8.504}$
- Yields within $\frac{b}{B_\alpha} = \frac{18}{127} = \frac{1}{7.056}$

Variance estimation

- For PPS, $n=ab$ is fixed. Hence,

$$\bar{y} = \frac{\sum_{\alpha=1}^a \sum_{\beta=1}^b y_{\alpha\beta}}{ab}$$

$$\text{var}(\bar{y}) = \frac{(1-f)}{a} s_a^2$$

Unit 6: Unequal Sized Clusters

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Unit 6: Unequal Sized Clusters

PPS

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C. Systematic PPS

- Compute zone size: $N/a = 2160/2 = 1080$
 - Select one subsample from each zone size 1080
 - Zone boundary falls 348 "elements" (measure units) through Unit 4; falls in 2 zones
- Select Units by selecting RS from 1 to 1080
 - Identify selection from cumulative counts
 - Add the interval $k = 1080$ to the RS
 - Identify next selected cluster
 - Select one subsample from each at the rate b/B_α

Selecting clusters with PPS

Unit	B_α	Cum. B_α
1	443	443
2	162	605
3	127	732
4	554	1286
5	115	1401

Unit	B_α	Cum. B_α
6	291	1692
7	64	1756
8	70	1826
9	232	2058
10	102	2160

Within sampling rates

- For example, suppose $RN = 804$
 - Select $\alpha = 4$
 - $RN + I = 804 + 1080 = 1884$ selects $\alpha = 9$
- For $\alpha = 4$, $\frac{aB_\alpha}{N} = \frac{B_\alpha}{N/a} = \frac{554}{2160/2} = \frac{554}{1080} = \frac{1}{1.950}$
- Subsample $\frac{b}{B_4} = \frac{18}{554} = \frac{1}{30.78}$

PPS without replacement

- Large literature on without replacement PPS selection procedures
- Alternatives lead to variance estimation expressions that require knowledge about the joint probabilities of selection
- Horvitz-Thompson estimator and the Yates-Grundy estimator of variance
- Unfortunately, the Yates-Grundy estimator can be negative
- Literature proposes wide variety of designs to overcome this problem

Unit 6: Unequal Sized Clusters

PPeS

- A. Estimated size measures
- B. Stratification
- C. Oversize units
- D. Undersize units

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A. Estimated size measures

- Suppose that the exact count of elements -- B_α -- in each cluster is unknown
- But measure that approximates the number of units (that is, the size) in each cluster -- Measure of Size -- Mos_α
- Example: Do not know the current exact count of employees for each Unit
 - Do know the number counted for each Unit at the last payroll one month ago

Example

Unit	Last payroll	Now
1	443	460
2	162	172
3	127	130
4	554	554
5	115	125
6	291	310
7	64	68
8	70	74
9	232	246
10	102	141
Total	2160	2280

- Probabilities proportionate to estimated size: *epsem* design in two stages
- Over-sample large clusters relative to small
- Select clusters with the rate $\frac{aMos_{\alpha}}{\sum_{\alpha=1}^A Mos_{\alpha}}$
- Then subsample elements at rate $\frac{b^*}{Mos_{\alpha}}$
- Why b^* ?
 - The actual number of second stage units selected is unknown; it's a target subsample size
 - If $Mos_{\alpha} = B_{\alpha}$, then $b^* = b$

Selecting units with PPeS

- Returning to employees example, suppose $b^*=18$, $a=2$, and
- $a = 2$ clusters are selected $\sum_{\alpha=1}^{10} Mos_{\alpha} = 2160$
- One subsample of expected size $b^* = 18$ each

$$f = \frac{2Mos_{\alpha}}{\sum Mos_{\alpha}} \times \frac{18}{Mos_{\alpha}} = \frac{36}{2160} = \frac{1}{60}$$

- With PPeS sampling, subsampling at a specified rate and not selecting a fixed number of elements from each selected cluster

Example

Unit	Last payroll	Now
1	443	460
2	162	172
3	127	130
4	554	554
5	115	125
6	291	310
7	64	68
8	70	74
9	232	246
10	102	141
Total	2160	2280

PPeS illustration

- Suppose $RN_1 = 702$ $RN_2 = 1744$

- Select Units 3 and 7

- For Unit 3,

$$\frac{b}{Mos_3} = \frac{18}{127} = \frac{1}{7.056}$$

- For Unit 7,

$$\frac{b}{Mos_7} = \frac{18}{64} = \frac{1}{3.556}$$

- If $Mos_\alpha = B_\alpha$, a sample of exactly size 36 will be selected

Sample size variation returns

- But here $Mos_{\alpha} \cong B_{\alpha}$
- For Unit 3, $B_{\alpha} = 130$, and $\frac{b}{Mos_3} = \frac{1}{7.056}$ yields

$$x_a = \frac{b}{Mos_{\alpha}} \times B_{\alpha} = \frac{1}{7.056} \times 130 = 18.425$$

- For interval 7.056, select a sample of **18** employees with probability 0.575
- Or a sample of **19** with probability 0.425

Unit 6: Unequal Sized Clusters

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Unit 6: Unequal Sized Clusters

PPeS

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B. Stratification

- Mos , a , b^* , etc., are specified within strata

$$f = f_h = \frac{a_h Mos_{h\alpha}}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}} \times \frac{b_h^*}{Mos_{h\alpha}} = \frac{a_h b_h^*}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}}$$

- Retain *epsem* for stratified PPeS sampling across strata: $f = f_h$ for all h .

Implicit stratification

- Systematic PPeS sampling implicitly stratifies by selecting within each zone one subsample size b^*
- Stratification notation not necessary with this design
- Zone size (or sampling interval) is

$$\frac{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}}{a_h}$$

Select $a = 4$ units

- $a = 4$ subsamples of size $b^* = 18$ from $Mos_t = 2160$

- Then

$$f = \frac{ab^*}{\sum Mos_{\alpha}} = \frac{4 \times 18}{2160} = \frac{1}{30}$$

- Zone size is $2160/4 = 540$

Explicit stratification

Stratum I		Stratum II	
Unit	Mos	Unit	Mos
1	443	5	115
2	162	6	291
3	127	7	64
4	554	8	70
		9	232
		10	102
Total	1286		874

Paired selection

$$f_1 = \frac{1}{30} \neq \frac{2Mos_{1\alpha}}{\sum_{\alpha=1}^{A_1} Mos_{1\alpha}} \frac{18}{Mos_{1\alpha}} = \frac{36}{1286}$$

$$f_2 = \frac{1}{30} \neq \frac{2Mos_{2\alpha}}{\sum_{\alpha=1}^{A_2} Mos_{2\alpha}} \frac{18}{Mos_{2\alpha}} = \frac{36}{874}$$

- Can manipulate $a_h, b_h^*, \sum_{\alpha=1}^{A_h} Mos_{h\alpha}$
- Uncommon to vary a_h

Vary subsample size

- For paired selections,

$$f_h = \frac{2Mos_{h\alpha}}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}} \frac{b_h^*}{Mos_{h\alpha}}$$

- Adjusting the subsample sizes across strata,

$$f_1 = \frac{2Mos_{1\alpha}}{\sum_{\alpha=1}^{A_1} Mos_{1\alpha}} \frac{b_1^*}{Mos_{1\alpha}} = \frac{1}{30}$$

$$b_1^* = 21.43$$

$$b_2^* = 14.57$$

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Unit 6: Unequal Sized Clusters

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- D. Undersize units

C. Oversize units

- Oversize units are clusters for which $Mos_{h\alpha}$ is so large that the unit has a certain chance of selection
 - Often so large that they will be selected multiple times in systematic PPeS
- For example, Unit 4 has $Mos_4 = 554 > 540$
 - Selected twice with probability

$$\frac{Mos_4}{540} = 1.0259$$

Solutions

- If there are few such clusters and chances of multiple selections small leave them in the list
 - If one selected k times, select k subsamples
- If there are many such clusters comprising a large share of the population, place in separate strata
 - Each such cluster is now a stratum, a cluster selected with certainty
 - Sampling rate(s) are applied directly within clusters
 - Self-representing units (SRU)

Unit 6: Unequal Sized Clusters

PPeS

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Unit 6: Unequal Sized Clusters

PPeS

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D. Undersize units

- Undersize units are clusters for which $Mos_{h\alpha} < cb^*$
- $Mos_{h\alpha} < b^*$ implies sampling within cluster at rate > 1
- All elements must be selected in the cluster, and

$$f = \frac{a_h Mos_{h\alpha}}{\sum Mos_{h\alpha}} \times 1$$

Solutions

- Link undersize units to form linked units of minimum sufficient size
 - Link before selection for the entire frame, especially if the frame is a computerized list
 - Can create clusters with greater heterogeneity
 - In area sampling, link geographically contiguous units
- Linking after selection
- If numerous, place in separate stratum
- Undersize units can include zero measure clusters

Linking after (during) selection

- (1) Identify selected unit (block). If the selected unit and next on the list are minimum sufficient size, **STOP**
- (2) If selected unit or next are not of minimum sufficient size,
 - a) **Move forward** in list until first unit of minimum sufficient size is encountered
 - b) **Cumulate units backwards** until a linked unit of minimum sufficient size is created
 - c) Continue process until the selected unit is linked

Example

Tract	Block	HU's
28	210	23
	211	17
	212	36
	213	1
29	101	0
	102	0
	103	0
	104	0
	105	1
	106	0
	107	0
	108	0
	109	0
	110	0
	111	0
	112	0
	113	0
	114	60
	115	9

Example

Tract	Block	HU's
28	210	23
	211	17
	212	36
	213	1
29	101	0
	102	0
	103	0
	104	0
	105	1
	106	0
	107	0
	108	0
	109	0
	110	0
	111	0
	112	0
	113	0
	114	60
	115	9

Link 101-113, 212-213, size = 38, e = 2

Link 105-113, size = 1

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- More control over sample size. Why?
 - Bias of r
 - Variance of r
- Equalize workloads controlling survey costs

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- A sample of $n = 36$ employees from $N = 2,160$
 - $f = 36/2160 = 1/60$

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Sample size control

- Redefine clusters
- Stratification by size, varying sampling rates by size of cluster stratum
- Give up *epsem* for equal size subsamples
- Probabilities proportionate to size (PPS)

Redefine clusters

- Create 10 clusters of approximately $N/10 = 216$ employees each
- Variation above and below desired cluster size
- Once clusters created, select the sample with uniform sampling fractions

$$f = f_a f_b = \left(\frac{a}{A}\right)\left(\frac{b}{B}\right)$$

$$fN = \left(\frac{1}{60}\right)(2160) = 36$$

Stratify by size

Stratum I		Stratum II	
Unit	No.	Unit	No.
2	162	1	443
3	127	4	554
5	115	6	291
7	64	9	<u>232</u>
8	70		1520
10	<u>102</u>		
	640		

Varying sampling rates

- Obtain *epsem* by choice of first and second stage rates:
- Use in stratum I: $f_{Ia} = \frac{1}{6}$ $f_{Ib} = \frac{1}{10}$
- Use in stratum II: $f_{IIa} = \frac{1}{4}$ $f_{IIb} = \frac{1}{15}$
- Sample size in stratum I ranges from 6 to 17, stratum II from 15 to 37
- $cv(x)$ has been reduced within strata.

Equal subsample size

- Select clusters with equal rates but take equal size subsamples
- Non-*epsem* design

$$f = f_a f_{b|a} = \left(\frac{a}{A}\right)\left(\frac{b}{B_\alpha}\right) = \left(\frac{2}{10}\right)\left(\frac{18}{B_\alpha}\right) = \frac{16}{10 \times B_\alpha} = f_\alpha$$

- Knowing B_α for each selected cluster, create weights $w_{\alpha\beta} = f_\alpha^{-1}$
- Constant overall workloads at the price of weighting and increases in sampling variances relative to *epsem* designs

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Unit 6: Unequal Sized Clusters

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PPS, or how to achieve *epsem* and equal subsample sizes at the same time

- Goal: Select a two stage sample such that there are equal subsample sizes in each cluster with *epsem*
- We want an overall *epsem* rate $f = n/N$ such that the sampling rate in the second stage is b/B_α

$$f = f_\alpha f_{\beta|\alpha} = \left(f_\alpha \right) \left(\frac{b}{B_\alpha} \right) = \left(\frac{n}{N} \right)$$

- Solving $f_\alpha = \frac{n}{N} \times \frac{B_\alpha}{b} = \frac{ab}{\sum_{\alpha=1}^A B_\alpha} \times \frac{B_\alpha}{b} = \frac{aB_\alpha}{\sum_{\alpha=1}^A B_\alpha}$

PPS implications

- Select clusters with probabilities proportionate to their size, B_α
 - Overall the selection becomes $f = \frac{aB_\alpha}{\sum_{\alpha=1}^A B_\alpha} \times \frac{b}{B_\alpha}$
- Selection probability varies at both first & second stages
 - Overall, *epsem*
 - Large clusters selected with **high probabilities**, *subsampling* at **low rate**
 - Small clusters selected with **low probabilities**, *subsampling* at **high rate**

Selecting clusters with PPS

- Two approaches
 - Simple replicated subsampling – at random and with replacement
 - Systematic sample and without replacement
- For both approaches, need to cumulate (sequentially add up) the size of the clusters

Selecting clusters with PPS

Unit	B_α	Cum. B_α	Unit	B_α	Cum. B_α
1	443	443	6	291	1692
2	162	605	7	64	1756
3	127	732	8	70	1826
4	554	1286	9	232	2058
5	115	1401	10	102	2160

Using cumulative counts

- Simple replicated subsampling for $a=2$:
 - Select RN from 1 to 2160 with replacement
 - Find the cluster (Unit) with cumulative sum first exceeding RN
 - Select
 - Repeat
 - Subsampling rate b / B_α requires exactly b
- If cluster selected k times, take $k \times b$ elements
- Subsampling: SRS, systematic, stratified element, even cluster sampling

Calculating subsampling rates

- For example, $RN_1=702$ and $RN_2 = 1744$ select Units 3 and 7
- For Unit 3, $\frac{aB_\alpha}{N} = \frac{2 \times 127}{2160} = \frac{1}{8.504}$
- Yields within $\frac{b}{B_\alpha} = \frac{18}{127} = \frac{1}{7.056}$

Variance estimation

- For PPS, $n=ab$ is fixed. Hence,

$$\bar{y} = \frac{\sum_{\alpha=1}^a \sum_{\beta=1}^b y_{\alpha\beta}}{ab}$$

$$\text{var}(\bar{y}) = \frac{(1-f)}{a} s_a^2$$

Unit 6: Unequal Sized Clusters

PPS

- A. Controlling sample size
- B. PPS
- C. Systematic PPS

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- Compute zone size: $N/a = 2160/2 = 1080$
 - Select one subsample from each zone size 1080
 - Zone boundary falls 348 "elements" (measure units) through Unit 4; falls in 2 zones
- Select Units by selecting RS from 1 to 1080
 - Identify selection from cumulative counts
 - Add the interval $k = 1080$ to the RS
 - Identify next selected cluster
 - Select one subsample from each at the rate b/B_α

Selecting clusters with PPS

Unit	B_{α}	Cum. B_{α}	Unit	B_{α}	Cum. B_{α}
1	443	443	6	291	1692
2	162	605	7	64	1756
3	127	732	8	70	1826
4	554	1286	9	232	2058
5	115	1401	10	102	2160

Within sampling rates

- For example, suppose $RN = 804$
 - Select $\alpha = 4$
 - $RN + I = 804 + 1080 = 1884$ selects $\alpha = 9$
- For $\alpha = 4$, $\frac{aB_\alpha}{N} = \frac{B_\alpha}{N/a} = \frac{554}{2160/2} = \frac{554}{1080} = \frac{1}{1.950}$
- Subsample $\frac{b}{B_4} = \frac{18}{554} = \frac{1}{30.78}$

PPS without replacement

- Large literature on without replacement PPS selection procedures
- Alternatives lead to variance estimation expressions that require knowledge about the joint probabilities of selection
- Horvitz-Thompson estimator and the Yates-Grundy estimator of variance
- Unfortunately, the Yates-Grundy estimator can be negative
- Literature proposes wide variety of designs to overcome this problem

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- B. Stratification
- C. Oversize units
- D. Undersize units

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A. Estimated size measures

- Suppose that the exact count of elements -- B_α -- in each cluster is unknown
- But measure that approximates the number of units (that is, the size) in each cluster -- Measure of Size -- Mos_α
- Example: Do not know the current exact count of employees for each Unit
 - Do know the number counted for each Unit at the last payroll one month ago

Example

Unit	Last payroll	Now
1	443	460
2	162	172
3	127	130
4	554	554
5	115	125
6	291	310
7	64	68
8	70	74
9	232	246
10	102	141
Total	2160	2280

- Probabilities proportionate to estimated size: *epsem* design in two stages
- Over-sample large clusters relative to small
- Select clusters with the rate $\frac{aMos_{\alpha}}{\sum_{\alpha=1}^A Mos_{\alpha}}$
- Then subsample elements at rate $\frac{b^*}{Mos_{\alpha}}$
- Why b^* ?
 - The actual number of second stage units selected is unknown; it's a target subsample size
 - If $Mos_{\alpha} = B_{\alpha}$, then $b^* = b$

Selecting units with PPeS

- Returning to employees example, suppose $b^*=18$, $a=2$, and
- $a = 2$ clusters are selected $\sum_{\alpha=1}^{10} Mos_{\alpha} = 2160$
- One subsample of expected size $b^* = 18$ each

$$f = \frac{2Mos_{\alpha}}{\sum Mos_{\alpha}} \times \frac{18}{Mos_{\alpha}} = \frac{36}{2160} = \frac{1}{60}$$

- With PPeS sampling, subsampling at a specified rate and not selecting a fixed number of elements from each selected cluster

Example

Unit	Last payroll	Now
1	443	460
2	162	172
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PPeS illustration

- Suppose $RN_1 = 702$ $RN_2 = 1744$

- Select Units 3 and 7

- For Unit 3,

$$\frac{b}{Mos_3} = \frac{18}{127} = \frac{1}{7.056}$$

- For Unit 7,

$$\frac{b}{Mos_7} = \frac{18}{64} = \frac{1}{3.556}$$

- If $Mos_\alpha = B_\alpha$, a sample of exactly size 36 will be selected

Sample size variation returns

- But here $Mos_{\alpha} \cong B_{\alpha}$
- For Unit 3, $B_{\alpha} = 130$, and $\frac{b}{Mos_3} = \frac{1}{7.056}$ yields

$$x_a = \frac{b}{Mos_{\alpha}} \times B_{\alpha} = \frac{1}{7.056} \times 130 = 18.425$$

- For interval 7.056, select a sample of **18** employees with probability 0.575
- Or a sample of **19** with probability 0.425

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B. Stratification

- Mos , a , b^* , etc., are specified within strata

$$f = f_h = \frac{a_h Mos_{h\alpha}}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}} \times \frac{b_h^*}{Mos_{h\alpha}} = \frac{a_h b_h^*}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}}$$

- Retain *epsem* for stratified PPeS sampling across strata: $f = f_h$ for all h .

Implicit stratification

- Systematic PPeS sampling implicitly stratifies by selecting within each zone one subsample size b^*
- Stratification notation not necessary with this design
- Zone size (or sampling interval) is

$$\frac{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}}{a_h}$$

Select $a = 4$ units

- $a = 4$ subsamples of size $b^* = 18$ from $Mos_t = 2160$

- Then

$$f = \frac{ab^*}{\sum Mos_{\alpha}} = \frac{4 \times 18}{2160} = \frac{1}{30}$$

- Zone size is $2160/4 = 540$

Explicit stratification

Stratum I		Stratum II	
Unit	Mos	Unit	Mos
1	443	5	115
2	162	6	291
3	127	7	64
4	554	8	70
		9	232
		10	102
Total	1286		874

Paired selection

$$f_1 = \frac{1}{30} \neq \frac{2Mos_{1\alpha}}{\sum_{\alpha=1}^{A_1} Mos_{1\alpha}} \frac{18}{Mos_{1\alpha}} = \frac{36}{1286}$$

$$f_2 = \frac{1}{30} \neq \frac{2Mos_{2\alpha}}{\sum_{\alpha=1}^{A_2} Mos_{2\alpha}} \frac{18}{Mos_{2\alpha}} = \frac{36}{874}$$

- Can manipulate $a_h, b_h^*, \sum_{\alpha=1}^{A_h} Mos_{h\alpha}$
- Uncommon to vary a_h

Vary subsample size

- For paired selections,

$$f_h = \frac{2Mos_{h\alpha}}{\sum_{\alpha=1}^{A_h} Mos_{h\alpha}} \frac{b_h^*}{Mos_{h\alpha}}$$

- Adjusting the subsample sizes across strata,

$$f_1 = \frac{2Mos_{1\alpha}}{\sum_{\alpha=1}^{A_1} Mos_{1\alpha}} \frac{b_1^*}{Mos_{1\alpha}} = \frac{1}{30}$$

$$b_1^* = 21.43$$

$$b_2^* = 14.57$$

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C. Oversize units

- Oversize units are clusters for which $Mos_{h\alpha}$ is so large that the unit has a certain chance of selection
 - Often so large that they will be selected multiple times in systematic PPeS
- For example, Unit 4 has $Mos_4 = 554 > 540$
 - Selected twice with probability

$$\frac{Mos_4}{540} = 1.0259$$

Solutions

- If there are few such clusters and chances of multiple selections small leave them in the list
 - If one selected k times, select k subsamples
- If there are many such clusters comprising a large share of the population, place in separate strata
 - Each such cluster is now a stratum, a cluster selected with certainty
 - Sampling rate(s) are applied directly within clusters
 - Self-representing units (SRU)

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D. Undersize units

- Undersize units are clusters for which $Mos_{h\alpha} < cb^*$
- $Mos_{h\alpha} < b^*$ implies sampling within cluster at rate > 1
- All elements must be selected in the cluster, and

$$f = \frac{a_h Mos_{h\alpha}}{\sum Mos_{h\alpha}} \times 1$$

Solutions

- Link undersize units to form linked units of minimum sufficient size
 - Link before selection for the entire frame, especially if the frame is a computerized list
 - Can create clusters with greater heterogeneity
 - In area sampling, link geographically contiguous units
- Linking after selection
- If numerous, place in separate stratum
- Undersize units can include zero measure clusters

Linking after (during) selection

- (1) Identify selected unit (block). If the selected unit and next on the list are minimum sufficient size, **STOP**
- (2) If selected unit or next are not of minimum sufficient size,
 - a) **Move forward** in list until first unit of minimum sufficient size is encountered
 - b) **Cumulate units backwards** until a linked unit of minimum sufficient size is created
 - c) Continue process until the selected unit is linked

Example

Tract	Block	HU's
28	210	23
	211	17
	212	36
	213	1
29	101	0
	102	0
	103	0
	104	0
	105	1
	106	0
	107	0
	108	0
	109	0
	110	0
	111	0
	112	0
	113	0
	114	60
	115	9

Example

Tract	Block	HU's
28	210	23
	211	17
	212	36
	213	1
29	101	0
	102	0
	103	0
	104	0
	105	1
	106	0
	107	0
	108	0
	109	0
	110	0
	111	0
	112	0
	113	0
	114	60
	115	9

Link 101-113, 212-213, size = 38, e = 2

Link 105-113, size = 1